

A Network Anatomy of Cross-Currency Basis Swap Spreads: Evidence from 25 Currencies

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Abstract

We develop a network-based framework where cross-currency basis swap spreads reflect currency- and tenor-specific inventory frictions, balance-sheet linkages, and global funding tightness. The model predicts: (i) dollar-pegged currencies exhibit muted volatility and rarely transmit shocks; (ii) longer-tenor bases are less volatile with weaker spillovers; (iii) crises amplify comovements; and (iv) spillover intensity is governed by structural centrality in the dealer-linkage network. Using a Bayesian graphical SVAR on 25 currencies—the largest sample to date—we uncover a sparse spillover network. Evidence supports all predictions: pegged legs remain insulated; longer maturities damp contagion; crises raise interconnectedness; transmission is euro-centered (CHF, DKK close behind).

Keywords: International Financial Markets, Covered Interest Parity, Cross-currency Basis Swap Spreads, Currency Swaps, Dollar Funding, Financial Crisis, Interconnectedness, BG-SVAR Model.

JEL: C11, C32, F31, G01, G15

1. Introduction

Persistent deviations from covered interest parity (CIP) have turned the cross-currency basis swap spreads into a state-dependent barometer of global dollar funding. Over the past two decades, the global financial system has experienced several significant disruptions, including the global financial crisis of 2007–2009, the European sovereign debt crisis of 2010–2013, and the COVID-19 pandemic in 2020. Beyond causing widespread financial and macroeconomic turmoil, these events disrupted long-established and seemingly robust relationships in economics and finance (Cerutti and Zhou, 2024; Eguren-Martin et al., 2024), thereby further propagating adverse economic and financial shocks within the system. By analysing the connections among major funding currencies, this paper examines how market interdependence

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and financial-market fragmentation are deteriorating, expressed as prolonged and pronounced deviations from the long-established and theoretically convenient idea of covered interest parity. Using network analysis and currency-interaction to investigate financial-market fragmentation for foreign exchange is a novel approach and a significant contribution to the literature. Our paper furthermore adds to the discussion on deviations from covered interest parity, which have resulted in non-zero cross-currency basis swap spreads and the emergence of arbitrage opportunities in the foreign exchange and currency derivatives markets. This focus is particularly relevant, as major market episodes tend to increase tail risks in the distribution of financial outcomes and trigger shifts in the relationships among global macroeconomic variables (Abakah et al., 2024). These shifts, as our paper will show, contribute to persistent deviations from CIP and further adverse second-round propagation effects. While it is important to understand these dynamics in their own right, financial relationships and resulting financial-stability concerns have come to the forefront of monetary policy concerns, not least since the Global Financial Crisis in 2008 (Stein, 2012).

The cross-currency swaps market is one of the world’s largest and most liquid segments of the global derivatives market.¹ In a cross-currency swap contract, two parties exchange principal and interest payments in different currencies over a specified period (Baba and Packer, 2009b; Coffey et al., 2009; Baba and Packer, 2009a). At the start of the swap, the parties exchange an equivalent amount of principal in their respective currencies, and at the end of the swap term, they re-exchange the principal amounts at the same initial exchange rate. During the swap’s life, the parties also exchange periodic interest payments based on the swapped principal amounts. These interest payments can be at fixed or floating rates, as agreed in the swap contract. Institutions and investors use these financial instruments to

¹According to the Bank for International Settlements (BIS Triennial Survey, 2022), *foreign exchange (FX) swaps* account for roughly half of global FX turnover (about 51%), while *cross-currency swaps (CCS)* represent around 2% of daily transactions by volume. However, these two instruments serve complementary purposes within the same dollar funding ecosystem: FX swaps dominate at short maturities (typically up to 3 months) and provide short-term rollover funding, whereas CCS contracts, which are mostly floating–floating structures, intermediate medium- to long-term funding exposures (1–30 years) for banks, corporates, and sovereign issuers. As such, although CCS have smaller daily turnover, they represent the long-term funding arm of the same intermediation chain and account for multi-trillion-dollar notional amounts outstanding. Indeed, the BIS and IMF both describe cross-currency swaps as a *core segment* of global dollar funding markets. Hence, describing the CCS market as large and liquid is fully consistent with BIS conventions and the broader academic literature (see Avdjiev et al., 2019a; Du et al., 2018a).

hedge currency exposure (i.e., to protect against adverse exchange rate risk), obtain funding in different currencies, and make extra profits (pickups) across currencies.

For instance, in a cross-currency swap deal, a non-USD party (e.g., a EUR party) often seeks access to USD. Suppose two choices are available: 1) borrow USD directly from the cash market, or 2) undertake a cross-currency swap to obtain synthetic USD from the swap market by first issuing in EUR and then swapping to USD. If the cost of borrowing USD directly is C and the cost of obtaining USD through a swap (synthetic USD) is S , the difference between these two costs is the cross-currency basis swap spread, commonly referred to as the *basis*. This is mathematically expressed as $D = C - S$, where D denotes the basis.²

Covered Interest Parity (CIP) is a principle asserting that the cost of borrowing a currency directly should equal the cost of obtaining the same currency through a swap, leading to a zero or negligible cross-currency basis swap spread (i.e., $D = 0$). When the basis is non-zero, it creates arbitrage opportunities. A positive basis indicates that synthetic USD is cheaper than direct USD, while a negative basis implies a premium for USD funding, reflecting its relative importance during periods of stress when market participants are searching for liquidity.

Although the efficient market hypothesis suggests that such opportunities should be quickly arbitrated away, historical evidence shows persistent deviations. To gain insight into the evolution of the basis, Figure 1 displays the cross-currency basis swap (XCCY) against the USD averaged for G-10 countries. From 2000 to 2006, the average basis swap spreads were negligible, with CIP holding approximately. Significant deviations first emerged during

²Formally, following [Avdjiev et al. \(2019a\)](#), the n -year cross-currency basis between currency i and the U.S. dollar is defined as the deviation from covered interest parity:

$$(1 + y_{t,t+n}^{\$})^n = (1 + y_{t,t+n}^i + x_{t,t+n}^i)^n \frac{S_{it}}{F_{it,t+n}},$$

which in logs yields

$$x_{t,t+n}^i = y_{t,t+n}^{\$} - (y_{t,t+n}^i - \rho_{t,t+n}^i), \quad \rho_{t,t+n}^i \equiv \frac{1}{n} [\log(F_{it,t+n}) - \log(S_{it})].$$

Hence the basis measures the difference between the direct U.S.-dollar interest rate in the cash market $y_{t,t+n}^{\$}$ (our C) and the synthetic dollar rate implied by swapping foreign funding into dollars $y_{t,t+n}^i - \rho_{t,t+n}^i$ (our S):

$$D = C - S = y_{t,t+n}^{\$} - (y_{t,t+n}^i - \rho_{t,t+n}^i).$$

This is the same linearized form used by [Du et al. \(2018a\)](#) and adopted in BIS conventions for the Bloomberg-quoted cross-currency basis. An earlier version of this paper, in which the D-C expression was introduced, circulated as [Ahelegbeya and Ibhagui \(2022\)](#), DEM Working Paper No. 191, University of Pavia.

the 2007–2009 global financial crisis, challenging the long-standing belief in its nearly universal validity. During the 2010–2013 Eurozone crisis, while the average basis swap spreads widened again, the extent was somewhat less pronounced. Following the crisis, these deviations decreased slightly, but they expanded again between 2015 and 2018, a period marked by political events such as Brexit, economic shocks including stock market turbulence, particularly in China, and several oil price shocks.

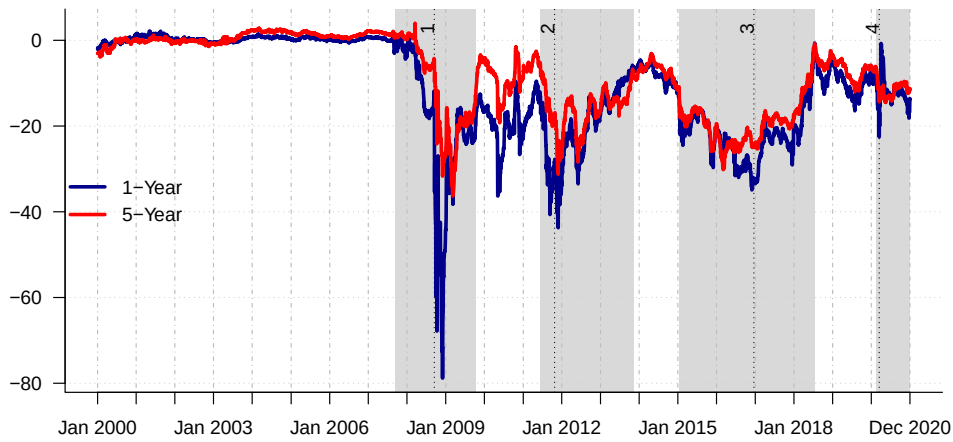


Figure 1: Averaged G-10 cross-currency basis swap spreads with 1- & 5-year maturities (Jan 2000 – December 2020). Grey-shaded regions correspond to 1) 2007–2009, 2) 2010–2013, 3) 2015–2018, and 4) 2020.

Studies on deviations from CIP and cross-currency basis swap spreads have spanned several decades. The first systematic examination of CIP deviations was [Callier \(1981\)](#), who proposed that transaction costs and market frictions could lead to minor, temporary deviations from CIP. These early studies established the foundation for understanding the circumstances under which CIP might not hold perfectly.

The global financial crisis of 2007–2009 marked a significant turning point in the study of CIP deviations. Pioneering analyses by [Baba et al. \(2008\)](#); [Baba and Packer \(2009b,a\)](#) documented substantial and persistent deviations from CIP during the crisis. They attributed these deviations to increased counterparty risk and shortages in USD liquidity, which disrupted the normal functioning of the cross-currency swap market.

Following the financial crisis, research expanded significantly to explore the underlying causes and persistence of CIP deviations. Studies by [Coffey et al. \(2009\)](#), [Fong et al. \(2010\)](#), and [Hui et al. \(2011\)](#) focused on regulatory changes and balance sheet constraints that con-

tinued to affect CIP. [Du and Schreger \(2016\)](#) highlighted that even at very short maturities, deviations from CIP provided non-trivial arbitrage opportunities, challenging classical limits-of-arbitrage models such as those proposed by [Shleifer and Vishny \(1997\)](#). Extending this analysis, [Du et al. \(2018b\)](#) found significant correlations between CIP deviations, other fixed income spreads, and nominal interest rates, offering deeper insights into the structural factors driving these deviations. [Ben Zeev and Nathan \(2024\)](#) emphasised that demand-side factors, particularly the impact of the Basel III accords and other macroprudential regulations, created frictions in financial intermediation efficiency, leading to long-lasting and significant changes in CIP. Finally, [Cunado et al. \(2024\)](#) noted, in a study spanning over 200 years, that the level of connectivity between financial markets fluctuates during major economic events and that the current breakdown of the no-arbitrage condition has historical precedents.

Several studies have highlighted the role of Federal Reserve interventions, such as extending dollar swap lines with global central banks, in mitigating CIP deviations. Notable contributions include [Baba and Packer \(2009a\)](#), [Aizenman and Pasricha \(2010\)](#), [Avdjiev et al. \(2020\)](#), [Stenfors et al. \(2022\)](#), and [Geyikci and Özyildirim \(2023\)](#). [Avdjiev et al. \(2019b\)](#) and [Cerutti et al. \(2021\)](#) examined macro-financial factors influencing CIP deviations, such as monetary policy divergences and global risk sentiment, demonstrating how broader economic and financial conditions impact the cross-currency basis swap spreads. [Ibhagui \(2019\)](#) showed that inflation differentials contribute to cross-currency basis swap spreads. Furthermore, [Aizenman et al. \(2022\)](#) argues that major central banks can alleviate acute strains in the offshore dollar funding market by providing liquidity through swap arrangements.

The persistent CIP deviations, along with the recent crisis caused by the pandemic, have sparked new debates among policymakers and led the Federal Reserve to take coordinated action. Understanding interconnectedness and spillover effects within the financial system is essential for informed policy decisions. For instance, it helps identify where interventions must be targeted, their impact on financial markets, and their broader macroeconomic implications.

Methodologically, network-based vector autoregressive (VAR) models have become increasingly popular for studying financial interconnectedness. [Ahelegbey et al. \(2016a,b\)](#) demonstrated the effectiveness of these models in examining contagion and spillovers in finan-

cial markets. Specifically, [Ahelegbey et al. \(2016a\)](#) employed the Bayesian graphical structural VAR (BGSVAR) approach to estimate direct spillovers within a sparse network framework, introducing a novel methodology for analyzing interconnected CIP deviations. [Cunado et al. \(2024\)](#) investigated the dynamic spillover of real interest rates across eight developed countries over the past two centuries using a time-varying VAR model enhanced by the connectedness methodology of [Balcilar et al. \(2021\)](#). [Chatziantoniou et al. \(2020\)](#) applied the same methodology to study event-dependent financial market connectedness, identifying interlinkages in balance sheets among G10 countries as a key factor in the persistence of CIP violations.

Our study extends the existing literature by addressing three central questions. (1) Do movements in the cross-currency basis swap spreads signal financial contagion? (2) Which currency plays a central role in the propagation of spillovers in the foreign exchange derivatives market? (3) What are the implications of these dynamics for financial market participants and policymakers? Using the BGSVAR model, we offer novel insights into CIP deviations and their interconnectedness during both tranquil and turbulent periods.

Our contributions are both theoretical and empirical. Firstly, we advance the literature on interconnected cross-currency basis swap spreads (XCCY) by identifying the central basis in the widening club and the additional profit/loss from hedging against adverse foreign exchange movements using XCCY. This extends and complements the study by [Wang et al. \(2024\)](#), who highlights the strength of the US dollar in shaping currency dependence. Secondly, we contribute to the literature on financial networks via network-VAR approximated models to study interconnectedness across hybrids of asset classes in financial markets ([Billio et al., 2012](#); [Diebold and Yilmaz, 2014](#); [Barigozzi and Hallin, 2017](#); [Ahelegbey et al., 2016a](#); [Billio et al., 2019](#)). Third, we add to the discussion about whether densely interconnected markets lower or increase the financial risks that come from shocks. In particular, we look at the currency derivatives markets ([Allen and Gale, 2000](#); [Freixas et al., 2000](#); [Billio et al., 2012](#); [Blume et al., 2013](#); [Haldane, 2013](#); [Acemoglu et al., 2015](#); [Chatziantoniou et al., 2020](#)).

The theoretical framework we develop is a dealer–intermediary model that microfound basis dynamics. In this setting, observed basis movements reflect currency- and tenor-specific inventory costs, cross-currency linkages arising from shared dealer balance sheets, and global

funding tightness that intensifies during crises. This model generates testable predictions on contagion, currency centrality, maturity heterogeneity, and pegged-currency stability, which we validate empirically.

Our dataset consists of daily Bloomberg data on 25 major funding currencies (the G-25) against the US dollar from January 2000 to December 2020. Employing a rolling BGSVAR, we uncover a sparse, directed network of contemporaneous and dynamic spillovers, while also tracking network density and a turbulence index over time. The results reveal that CIP deviations are tightly linked across currencies, particularly during crises. Interconnectedness is strongest and most persistent in turbulent periods. This means mark-to-market losses for investors with long basis exposures and larger arbitrage opportunities for new entrants.

Four main findings follow from this analysis. First, currencies tightly pegged to the US dollar display remarkably low XCCY volatility and are far less likely to transmit shocks, even during common stress episodes, because idiosyncratic dollar demand in the pegged leg is negligible and its balance-sheet linkages are weak.

Second, longer maturities tie up dealer balance sheets for longer and therefore carry higher inventory costs. This raises the *level* of the basis in the long leg but *reduces* its volatility and its outward spillovers. Under empirically grounded primitives—smaller and less-correlated demand at longer maturities and weaker cross-currency linkages with tenor—long-tenor bases are smoother and less contagious than short-tenor bases. The panel evidence confirms both the variance ordering and the attenuation of cross-currency co-movement.

Third, when global dollar funding tightens, the effective pricing kernel of the dealer sector becomes nearly one-dimensional. Cross-sectional interconnectedness expands, and the basis moves across currencies, becoming highly synchronised. This microfound the crisis-time ‘single-factor’ behaviour highlighted in prior work and matches the near-lockstep dynamics we observe in 2008–2009 and 2020.

Fourth, spillovers depend on balance-sheet linkages, with the Euro ranking as the dominant transmitter, followed by the Swiss franc and Danish krone, while the Japanese yen frequently acts as a receiver. These rankings remain stable across maturities and subsamples. Robustness checks, including alternative priors, rolling window lengths, subsamples, and sign

or scale normalisations, reinforce these findings.

Relative to the existing literature, which emphasizes persistent CIP deviations, dealer constraints, policy backstops, and dollar factors (e.g., [Baba and Packer, 2009b](#); [Coffey et al., 2009](#); [Du et al., 2018b](#); [Avdjiev et al., 2019b](#)), our study offers four new contributions: (i) We provide a microfoundation for crisis-limit co-movement via time-varying global funding tightness; (ii) develop a structural network metric that links currency centrality to dealer balance-sheets; (iii) establish a formal tenor ordering that reconciles higher long-tenor levels with lower volatility and weaker spillovers; and (iv) document new evidence on pegged-currency immunity and Euro-centered contagion using a broad G-25 panel at daily frequency.

The paper is organised as follows: Section 3 presents the Bayesian network-VAR model, Section 4 describes the data, Section 5 reports the results, and Section 7 concludes the paper.

2. Theoretical Foundations

To formalise the dynamics we observe in the market, we develop a theoretical framework to explain the interconnected dynamics of cross-currency basis swap spreads. The model builds directly on the idea of strong contagion effects, centrality of core currencies and heterogeneity across pegged and commodity-linked currencies. Our goal is to microfound these dynamics within a tractable asset-pricing environment that incorporates intermediary frictions and heterogeneous client demand.

2.1. Model Setup

We consider n currencies indexed by $k \in \{1, \dots, n\}$. We work in a discrete time framework, so that $t = 0, 1, 2, \dots$. For each currency k , define $b_{k,t}$ as the deviation from covered interest parity (CIP), i.e. the cross-currency basis swap spread at time t . We stack the spreads associated with the currencies in the vector $b_t = (b_{1,t}, \dots, b_{n,t})'$.

In each period, global clients demand USD funding in exchange for foreign currency k . We denote this net demand by $d_{k,t}$, where $d_{k,t} > 0$ indicates that clients require USD, and $d_{k,t} < 0$ indicates that they supply USD. Dealers stand on the opposite side of these transactions, absorbing the client positions and providing the required USD. To do so, they source USD

through various channels, incurring costs in the process. These costs are passed on to clients in the form of a wider basis: the greater the dealer’s marginal cost of sourcing USD, the larger the corresponding increase in the basis.

Let $d_t = (d_{1,t}, \dots, d_{n,t})'$ denote the vector of global clients’ net USD demand across currencies $k \in \{1, \dots, n\}$, and let $q_t = (q_{1,t}, \dots, q_{n,t})'$ denote the vector of dealers’ USD supply. Equilibrium in the client-dealer transaction requires that

$$d_{k,t} = q_{k,t}, \quad \text{for all } k \in \{1, \dots, n\} \quad (1)$$

2.2. Dealer Costs

Dealers face convex costs of absorbing positions due to balance sheet frictions, risk management constraints, and regulatory capital charges.

Following [He and Krishnamurthy \(2013\)](#) and [Garleanu and Pedersen \(2011\)](#), we model dealers as absorbing client order flow subject to convex balance-sheet costs. This specification also builds on the inventory cost tradition of [Ho and Stoll \(1981\)](#), where quadratic costs naturally arise as a reduced form of risk-management and capital constraints. In the FX basis context, [Du et al. \(2018b\)](#) and [Borio et al. \(2016\)](#) provide both theoretical and empirical support for viewing deviations from covered interest parity (CIP) as the price of these intermediary balance-sheet costs.

Accordingly, we include an explicit quadratic cost function that captures (i) maturity- and currency-specific inventory frictions, (ii) cross-currency linkages from shared balance sheets, and (iii) a global funding tightness parameter that rises during crises.

We therefore assume a quadratic cost function:

$$C_t(q_t) = \frac{1}{2} q_t' (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') q_t \quad (2)$$

where $T = \text{diag}(\tau_1, \dots, \tau_n)$ captures maturity- and currency-specific inventory costs, $\Phi = (\phi_{ij})$ is a symmetric matrix with $\phi_{ij} \geq 0$ that captures cross-currency linkages through shared dealer balance sheets, and $\Lambda_t \geq 0$ is a global funding tightness parameter, rising in crises. The rank-one term $\Lambda_t \mathbf{1}\mathbf{1}'$ generates co-movement across all currencies. The matrix $T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}'$

is positive semi-definite, ensuring convexity of the cost function.

2.3. Equilibrium Pricing

Dealers are risk-neutral and competitive. Zero-profit pricing implies that basis spreads equal the marginal cost of supplying USD. Using the condition $q_t = d_t$, we obtain

$$b_t = \nabla_{q_t} C_t(q_t) = (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') d_t \quad (3)$$

Thus, observed basis spreads are a linear transformation of client USD demand, with the pricing kernel determined by (i) idiosyncratic inventory frictions (T), (ii) cross-currency balance-sheet linkages (Φ), and (iii) systemic stress (Λ_t).

Interpretation of Volatility and Funding Frictions.. While the model allows volatility, tail risk, and uncertainty to enter as state variables, these are not treated as direct causes of the basis. Rather, they act as proximate determinants of the shadow cost of dollar funding, captured by the systemic-stress parameter Λ_t . Increases in market volatility or option-implied skew tighten dealers' balance-sheet constraints through higher margin requirements, liquidity haircuts, or capital charges, thereby raising Λ_t and widening the basis. Hence, volatility in this framework is a transmission variable that amplifies funding frictions, not an alternative explanation for them. This interpretation aligns with recent evidence that the cross-currency basis reflects frictions in wholesale funding markets—i.e., the shadow price of unencumbered dollar balance-sheet capacity—rather than pure price spillovers (Rime et al., 2022; Klok et al., 2024).

Interpretation of Cross-Currency Linkages.. The off-diagonal elements of the cost matrix $\Phi = (\phi_{ij})$ capture overlapping funding channels across currencies. A positive ϕ_{ij} means that warehousing exposures in currencies i and j draw on shared balance-sheet or collateral resources—such as dollar funding lines, internal capital limits, or repo haircuts—so that holding the two positions jointly is more costly than holding them separately. This structure produces heterogeneous spillovers: currencies with larger or denser ϕ_{ij} connections transmit funding stress more strongly. Empirically, this maps to dealer balance sheets that intermediate

multiple currency pairs through common dollar liquidity pools. This formulation provides a direct theoretical counterpart to the empirical spillover network estimated in empirical part of this paper, where significant directed edges correspond to nonzero ϕ_{ij} linkages.

2.4. Dealer Costs and Equilibrium Pricing

Following the dealer–intermediation literature (see [Duffie, 2010](#); [Garleanu and Pedersen, 2011](#); [He and Krishnamurthy, 2013](#)), we assume that intermediaries absorb client demand at a convex cost that reflects regulatory balance-sheet usage, inventory management, and systemic funding pressures. Let $q_t = (q_{1,t}, \dots, q_{n,t})'$ denote the vector of dealer positions across n currencies at time t . The total cost of warehousing these positions is specified as

$$C_t(q_t) = \frac{1}{2} q_t' (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') q_t \quad (4)$$

where $T = \text{diag}(\tau_1, \dots, \tau_n)$ is a diagonal matrix of maturity- and currency-specific inventory costs, $\Phi = (\phi_{ij})$ is a symmetric matrix with non-negative entries that capture cross-currency linkages, and $\Lambda_t \geq 0$ is a parameter representing global funding tightness.

A larger τ_k indicates that absorbing one additional unit of demand in currency k places greater strain on dealer balance sheets. Off-diagonal elements of Φ reflect the fact that warehousing positions in two currencies may use overlapping funding channels, so that holding them jointly is more costly than holding them in isolation. The global stress parameter Λ_t adds a common surcharge across all positions: in tranquil periods, this term is small, but in crisis periods it rises sharply, making costs highly correlated across currencies.

For convenience, we define

$$K_t \equiv T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}' \quad (5)$$

which summarises the full structure of balance-sheet costs faced by dealers at time t . The diagonal entries of K_t combine idiosyncratic inventory costs with systemic surcharges from Λ_t , determining the marginal expense of absorbing one more unit of demand in each currency. The off-diagonal entries reflect both cross-currency linkages and the global stress term, thereby

transmitting costs across markets and generating co-movement in basis spreads during crises.

Intuitively, K_t can be interpreted as the pricing kernel of the dealer sector. Because dealers are risk-neutral and competitive, zero-profit pricing implies that basis spreads equal marginal costs. Using the market-clearing condition $q_t = d_t$, we obtain the equilibrium pricing relation

$$b_t = K_t d_t \tag{6}$$

In normal times, K_t is dominated by T and Φ , so that spreads primarily reflect idiosyncratic inventory costs and bilateral linkages. During crises, Λ_t becomes large, and K_t approaches a near rank-one structure that forces strong co-movement across all basis spreads. This explains why CIP deviations, which appear fragmented in normal times, suddenly synchronise across markets when systemic funding pressures bind.

The quadratic form ensures convexity, so that marginal costs rise linearly with position size: larger positions become increasingly expensive to hold. This convex specification, standard in the intermediation literature, captures both risk-management constraints and regulatory capital charges.

2.5. Equilibrium Pricing

Dealers are assumed to be risk-neutral and perfectly competitive. Risk neutrality implies that they require no additional compensation for risk exposure beyond the cost of balance-sheet usage. Perfect competition implies zero expected profit: dealers set prices equal to marginal costs, as any markup would be competed away.

Formally, basis spreads $b_t = (b_{1,t}, \dots, b_{n,t})'$ are set equal to the gradient of the cost function with respect to dealer positions:

$$b_t = \nabla_{q_t} C_t(q_t) = (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') q_t \tag{7}$$

Since equilibrium requires $q_t = d_t$, with dealers absorbing the entire client demand vector,

this simplifies to

$$b_t = (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') d_t \quad (8)$$

This equilibrium condition conveys three key economic mechanisms. First, inventory frictions represented by T ensure that basis spreads rise with the intensity of demand in each currency, with larger effects in currencies or maturities with higher inventory costs. Second, cross-currency linkages captured by Φ transmit shocks across currencies: a large ϕ_{ij} indicates that demand pressure in currency i generates spillovers into currency j through shared balance-sheet constraints. Third, systemic stress captured by Λ_t produces global co-movement during crises; even idiosyncratic demand shocks can generate widespread widening of basis spreads when Λ_t rises, explaining the surge in cross-currency correlations observed empirically.

In short, this formulation shows that the cross-currency basis is not an arbitrary deviation from covered interest parity but reflects the shadow price of dealer balance-sheet usage. The matrix $(T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}')$ summarises the cost kernel of the intermediation sector, mapping raw client demands into observed basis spreads.

2.6. Testable Propositions

We now present testable propositions that anchor the theory and guide the empirical section. The model delivers several predictions, summarised below.

Proposition 1 (Contagion Amplification). *During crises, as Λ_t rises, interconnectedness of the cross-currency basis swap spreads increases, with correlations converging to unity in the limit as $\Lambda_t \rightarrow \infty$. See Appendix A.2 for formal derivation.*

This reflects the role of Λ_t as a global funding stress parameter. When dollar scarcity rises, the marginal cost of supplying USD increases uniformly across markets, effectively binding currencies through a common systemic factor. Basis spreads, therefore, become highly correlated in stress episodes. In the extreme, crises erase currency-specific idiosyncrasies, creating global synchronicity.

Proposition 2 (Currency Centrality). *Define the centrality of currency k in the network as*

$$C_k = \sum_{j \neq k} \phi_{kj} \quad (9)$$

A currency k is a central transmitter of shocks if C_k is maximal, i.e.

$$C_k > C_i \quad \forall i \neq k \quad (10)$$

In this case, shocks to the demand for currency k propagate most strongly to the rest of the system via $K_t = T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}'$. See Appendix A.3.

The matrix Φ encodes the network of cross-currency spillovers. If a currency k maintains strong financial linkages, through deep swap markets, regional ties, or global dealer usage, then ϕ_{kj} is large, and k functions as a systemic transmitter of shocks. Empirically, we find that the EUR, CHF, and DKK occupy such central roles.

Proposition 3 (Pegged Currency Immunity). *If currency k is tightly pegged to the USD, then $d_{k,t} \approx 0$, so $b_{k,t}$ remains muted even during crises. See Appendix A.4.*

Pegged currencies (such as HKD or QAR) generate little independent hedging demand, since their exchange rate risk relative to USD is contractually or institutionally suppressed. As a result, their basis spreads remain stable even when systemic shocks propagate elsewhere. This is consistent with their weak participation in contagion dynamics.

Proposition 4 (Maturity Heterogeneity). *If τ_k increases with maturity, then longer-tenor basis spreads are less volatile and exhibit weaker spillover dynamics than short-tenor spreads. See Appendix A.5.*

The diagonal element τ_k in $T = \text{diag}(\tau_1, \dots, \tau_n)$ represents currency- and tenor-specific inventory costs. Longer maturities tie up dealer balance sheets for extended horizons, which raises τ_k . In equilibrium,

$$b_t = (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') d_t \quad (11)$$

so short-tenor spreads, where τ_k is small, are dominated by cross-currency spillovers and systemic stress. This makes them highly sensitive to global shocks and tightly co-move across currencies. In contrast, at longer maturities, higher τ_k dampens global spillovers, yielding smoother and more idiosyncratic dynamics—a pattern that we confirm empirically.

Tenor Segmentation. Short-tenor basis spreads reflect temporary funding conditions and are thus more responsive to systemic stress Λ_t , whereas long-tenor spreads embed persistent capital costs and remain relatively inert to short-run shocks.

Interpretation of Tenor Effects.. While one might argue that dealer inventories respond endogenously to volatility and risk aversion, the cost function $C_t(q_t)$ in our framework already represents the equilibrium shadow price of balance-sheet usage after dealers have optimized their exposures. Hence, modeling an explicit dynamic inventory choice (à la Garleanu and Pedersen, 2011) would reduce to the same marginal-cost representation at equilibrium. Longer-tenor cross-currency swaps remain less sensitive to systemic stress not because dealers ignore risk, but because such positions are structurally costly to unwind and dominated by persistent capital and valuation-adjustment charges (KVA, FRTB). In contrast, short-tenor markets are margin-driven, rolled frequently, and thus transmit funding shocks more directly—producing stronger spillovers.

3. Bayesian Graphical SVAR Model

This section introduces the Network SVAR Model, which captures both temporal and contemporaneous dependencies among variables within a network structure. We then outline the Bayesian estimation procedure, which leverages sparsity-inducing priors and graphical modelling techniques for efficient estimation in high-dimensional settings.

3.1. Network SVAR Model

A network model represents relationships among variables using nodes and edges, offering insights into serial, temporal, and contemporaneous dependencies [Ahelegbey et al. \(2016a\)](#). Let $Y_t = (Y_{1,t}, \dots, Y_{n,t})$ be an n -dimensional vector of variables at time t . In a structural

vector autoregression (SVAR) framework, the model is given by:

$$Y_t = B_1 Y_{t-1} + \dots + B_p Y_{t-p} + U_t, \quad U_t \sim \mathcal{N}(0, \Sigma_u) \quad (12)$$

$$U_t = B_0 U_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon), \quad (13)$$

where B_s ($s = 1, \dots, p$) are lag-specific coefficient matrices, U_t denotes the reduced-form residuals, B_0 captures contemporaneous dependencies, and ε_t represents structural shocks.

The covariance of U_t relates to B_0 and Σ_ε as: $\Sigma_u = (I - B_0)^{-1} \Sigma_\varepsilon (I - B_0)^{-1'}$. To uncover contemporaneous linkages among currencies, we impose a sparse graphical prior on the precision matrix $\Omega = \Sigma_u^{-1}$. Zeros in Ω_{ij} imply conditional independence between currencies i and j given all others, defining the edges of the network. The resulting structure corresponds to a structural form $A_0 u_t = \varepsilon_t$ with independent structural shocks ε_t , but estimation proceeds directly in terms of Ω without requiring explicit identification of A_0 .

Let $B = (B_0, B_1, \dots, B_p)$ denote the stacked set of coefficient matrices, where $B_{ij,s}$ represents the impact of $Y_{j,t-s}$ on $Y_{i,t}$. To characterise the implied directional dependencies in (12), we define two types of adjacency matrices: the unweighted adjacency matrix A , which captures the existence of a connection from Y_j to Y_i , and the weighted adjacency matrix W , which captures the strength of that connection across all lags (including contemporaneous interactions). Formally, the (i, j) -th elements of these matrices are defined as:

$$A_{ij} = \begin{cases} 1, & \text{if } \sum_{s=0}^p |B_{ij,s}| > 0 \\ 0, & \text{otherwise} \end{cases} \implies Y_j \rightarrow Y_i \text{ if } A_{ij} = 1, \quad W_{ij} = \sum_{s=0}^p |B_{ij,s}|$$

A_{ij} reflects whether any statistically meaningful influence of Y_j on Y_i exists over the considered lags, while W_{ij} quantifies the total directional influence. The absolute value in the summation for A_{ij} avoids cancellations across lags and ensures a nonzero threshold for detection.

3.2. Bayesian Graphical VAR Model Estimation

To induce sparsity and enhance interpretability in VAR models, we follow the framework of [Ahelegbey et al. \(2016a\)](#), by introducing binary indicators $G_{ij,s} \in \{0, 1\}$, which determines

whether the coefficient $B_{ij,s}$ is active (non-zero) or excluded (zero), defined as:

$$G_{ij,s} = \begin{cases} 1 & \text{if } Y_{j,t-s} \rightarrow Y_{i,t} \implies B_{ij,s} \neq 0, \\ 0 & \text{if } Y_{j,t-s} \not\rightarrow Y_{i,t} \implies B_{ij,s} = 0. \end{cases}$$

We specify the prior distributions as follows:

$$\begin{aligned} [B_{ij,s} \mid G_{ij,s} = 1] &\sim \mathcal{N}(0, \eta), & G_{ij,s} &\sim \text{Ber}(\pi_{ij}), \\ \Sigma_u^{-1} &\sim \mathcal{W}(\delta_u, \Lambda_{u,0}), & \Sigma_\varepsilon^{-1} &\sim \mathcal{W}_{G_\varepsilon}(\delta_\varepsilon, \Lambda_{\varepsilon,0}) \end{aligned}$$

with hyperparameters $\eta, \pi_{ij}, \delta_u, \delta_\varepsilon, \Lambda_{u,0}$ and $\Lambda_{\varepsilon,0}$. When $G_{ij,s} = 1$, the coefficient $B_{ij,s}$ follows a Gaussian prior; if $G_{ij,s} = 0$, the coefficient is excluded, inducing sparsity. The reduced-form precision Σ_u^{-1} follows a Wishart distribution, while Σ_ε^{-1} is G-Wishart distributed over independence graph G_ε . Under the assumption that components of ε_t are mutually independent, G_ε is the empty graph (no off-diagonals), so Σ_ε^{-1} is diagonal.

We conduct posterior inference via a collapsed Gibbs sampler, cycling through:

$$\begin{aligned} &P(G_{1:p} \mid Y, p), \quad P(G_0 \mid Y, p, \hat{G}_{1:p}), \quad P(B_{1:p} \mid Y, p, \hat{G}_{1:p}, \Sigma_u), \\ &P(B_0 \mid Y, p, \hat{G}, \hat{B}_{1:p}, \Sigma_\varepsilon, \Sigma_u), \quad P(\Sigma_\varepsilon \mid Y, p, \hat{G}, \hat{B}, \Sigma_u), \quad P(\Sigma_u \mid Y, p, \hat{G}, \hat{B}, \hat{\Sigma}_\varepsilon). \end{aligned}$$

In our application, we set $\pi_{ij} = 0.5$, $\eta = 100$, $\delta_\varepsilon = \delta_u = n+2$, $\Lambda_{\varepsilon,0} = \delta_\varepsilon I_n$, $\Lambda_{u,0} = \delta_u I_n$. Convergence is assessed via the Gelman–Rubin statistic [Gelman and Rubin \(1992\)](#). Robustness is checked with out-of-sample validation and cross-validation. The sampler is implemented in MATLAB with code optimisation and parallelisation to handle high-dimensional systems.

4. Data Description

Existing studies on cross-currency basis swap spreads have mostly focused on the G-10 currencies. However, recent events have shown a shift in the ranking of currencies, with some G-10 members gradually losing their status as developed market currencies. Prominent among such currencies is the British pound (GBP), traditionally part of the G-5 currency

group, which has seen its status affected by events like Brexit, making it more akin to an emerging market currency. Conversely, some emerging market currencies³ gradually gained quasi-developed market status due to their resilience during events such as the pandemic.

Our research looks at how changes in covered interest parity (CIP) are linked. It uses daily data from Bloomberg from January 2000 to December 2020 and compares 25 major funding currencies (G-25) to the US dollar. Our G-25 consists of 13 developed and 12 emerging economies, following the Morgan Stanley Capital International Market Classification. See Table 1 for a description of the currencies.

Market	No.	Country/Region	Currency	XCCY	Start Date	End Date
Developed	1	Euro Area	Euro	EUR	3/1/2000	31/12/2020
	2	Japan	Japanese yen	JPY	3/1/2000	31/12/2020
	3	United Kingdom	British pound	GBP	3/1/2000	31/12/2020
	4	Australia	Australian dollar	AUD	3/1/2000	31/12/2020
	5	New Zealand	New Zealand dollar	NZD	3/1/2000	31/12/2020
	6	Canada	Canadian dollar	CAD	3/1/2000	31/12/2020
	7	Denmark	Danish krone	DKK	3/1/2000	31/12/2020
	8	Switzerland	Swiss franc	CHF	3/1/2000	31/12/2020
	9	Sweden	Swedish krona	SEK	3/1/2000	31/12/2020
	10	Norway	Norwegian krone	NOK	3/1/2000	31/12/2020
	11	Israel	Israeli shekel	ILS	3/7/2011	31/12/2020
	12	Hong Kong	Hong Kong dollar	HKD	3/1/2000	31/12/2020
	13	Singapore	Singapore dollar	SGD	3/1/2000	31/12/2020
Emerging	14	Thailand	Thai baht	THB	26/6/2006	31/12/2020
	15	Malaysia	Malaysian ringgit	MYR	9/5/2005	31/12/2020
	16	Korea	Korean won	KRW	26/2/2003	31/12/2020
	17	Russia	Russian ruble	RUB	2/11/2006	31/12/2020
	18	South Africa	South African rand	ZAR	3/1/2000	31/12/2020
	19	Turkey	Turkish lira	TRY	19/9/2006	31/12/2020
	20	UAE	UAE dirham	AED	14/8/2009	31/12/2020
	21	Saudi Arabia	Saudi riyal	SAR	27/11/2009	31/12/2020
	22	Chile	Chilean peso	CLP	24/4/2009	31/12/2020
	23	Mexico	Mexican peso	MXN	24/2/2000	31/12/2020
	24	China	Chinese yuan	CNY	21/7/2009	31/12/2020
	25	Qatar	Qatari riyal	QAR	27/8/2012	31/12/2020

Table 1: The major 25 world cross-currencies grouped according to the MSCI Market Classification.

In Figure 2, we report the time series of the XCCY for the G-25 currencies grouped according to developed and emerging markets with 1- and 5-year maturities. Apart from data limitations, several important reasons motivate our maturity choice. We focus on 1-year and 5-year maturities to capture medium- and long-term funding conditions beyond the

³See <https://www.refinitiv.com/perspectives/market-insights/how-hard-has-covid-19-hit-em-currencies/>

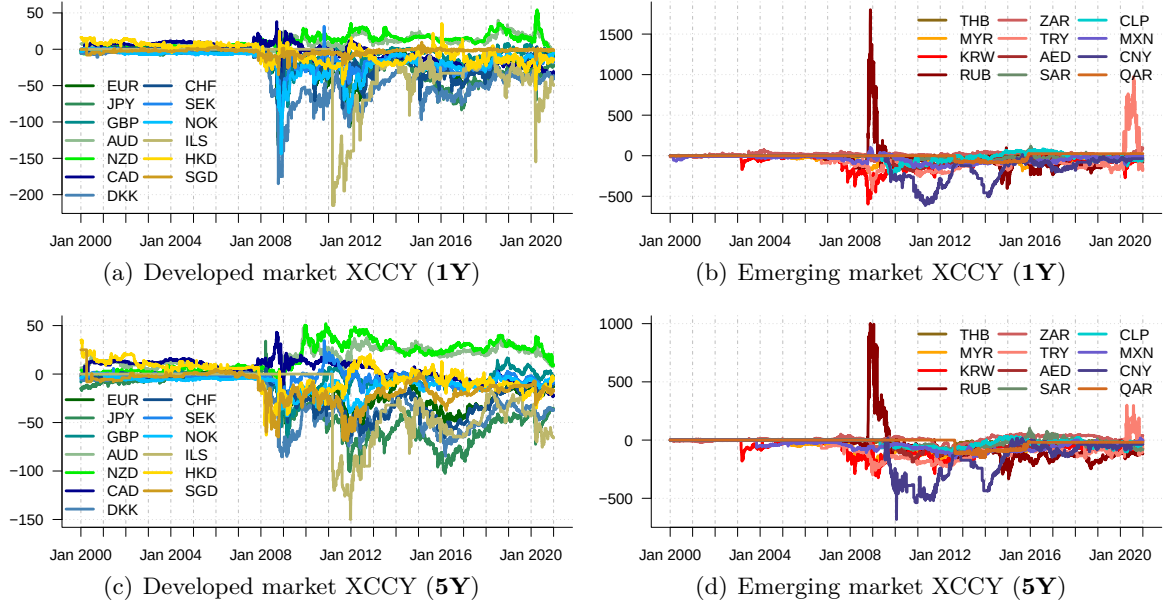


Figure 2: Time series of Cross-currency basis swap spreads (Jan 2000–December 2020).

overnight and 3-month segments. Shorter-tenor bases are often heavily affected by policy interventions, while longer maturities not only reflect policy interventions but also reflect structural balance-sheet costs and cross-currency term risk. Our choice follows the literature, e.g. Avdjiev et al. (2019) and Liao (2020), who use higher maturities which highlight that CIP deviations at 1-year and beyond embed persistent balance-sheet and risk-transfer components rather than more transient liquidity spreads as seen in the front end of the basis curve.

We notice that between 2000 and 2006, most of the basis spreads were generally within the neighbourhood of zero. This suggests that in the years preceding the global financial crisis (GFC), any CIP rule held approximately, and any deviation was quickly arbitrated away because USD was more readily available to market participants. The supply of dollar funding was hardly ever constrained, as there was the absence or diminished presence of any financial intermediation channel, and neither dollar funding specialness nor foreign currency specialness existed.

During the crisis of 2007–2009, the XCCY showed significant pickups in the CIP deviations, most of which were negative. During this period, the supply of dollar funding was highly constrained, an intense dollar funding specialness emerged, financial intermediaries

became more constrained, and a strong presence of a financial intermediation channel arose.

The timeline of events shows that during the GFC, non-US entities (foreign financial institutions with interests in the US) needed USD to fund their US operations. However, these institutions were unable to access the USD cash market directly because US financial institutions were concerned about counterparty risk. As a result, these non-US entities turned to the cross-currency swap market, where they swapped non-USD currency for USD. This raised the cost of synthetic US dollars in the swap market due to the high demand for dollar hedges relative to the supply of dollar hedges. The higher cost of synthetic dollars in cross-currency swap markets forced foreign entities to pay a premium (compared to the cost of raising direct USD in the cash market), which led to what we know today as cross-currency basis swap spreads. In 2010–2013, during the Eurozone crisis, the basis spreads widened again, albeit to a lesser extent compared to the GFC.

Table 2 presents the statistics of the basis spreads over three crisis periods: 2007–2009, 2010–2013, and 2020. According to the table, the most volatile basis during the three crisis periods is usually from emerging market currencies. For both 1- and 5-year maturity basis, the Russian ruble (RUB) recorded the highest standard deviation of 321.6 between 2007–2009, the Chinese yuan (CNY) recorded the highest volatility between 2010–2013, and the Turkish lira (TRY) topped the list of the most volatile in 2020. Among the developed market currencies, the Danish krone (DKK) was the most volatile in 2007–2009, while the Israeli shekel (ILS) topped the list between 2010–2013 and 2020.

The basis can be positive or negative as seen from Figure 2 and the statistics in Table 2. Among our G-25 currencies, the majority recorded negative bases for the 1- and 5-year maturities except for the Australian dollar (AUD) and the New Zealand dollar (NZD). These exceptions reported positive basis spreads for both maturities.

During the crisis periods shown in Table 2, as well as in the pre-and post-crisis periods illustrated in Figure 2, the basis for the AUD and NZD remained positive even when others became strongly negative. This implies that while other non-US entities were paying a premium to obtain (synthetic) USD in the swap market, the AUD and NZD entities could obtain synthetic USD at a discount compared to the cost of direct USD in the cash market.

	1-Year Maturity XCCY			5-Year Maturity XCCY		
	2007–2009	2010–2013	2020	2007–2009	2010–2013	2020
EUR	-21.4 (23.0)	-35.7 (19.9)	-11.1 (4.2)	-14.8 (16.2)	-29.6 (11.9)	-16.6 (4.0)
JPY	-14.2 (15.1)	-28.0 (9.4)	-29.9 (9.6)	-13.5 (20.1)	-58.5 (15.4)	-46.1 (8.4)
GBP	-20.8 (22.4)	-8.5 (7.1)	-2.3 (3.4)	-17.1 (18.1)	-7.1 (7.6)	-0.4 (3.3)
AUD	6.2 (9.1)	10.8 (6.3)	13.1 (10.8)	9.3 (11.8)	25.0 (6.3)	17.7 (5.9)
NZD	9.5 (5.2)	18.5 (5.9)	13.5 (15.5)	7.9 (8.7)	36.1 (6.3)	22.4 (8.1)
CAD	7.1 (7.8)	-3.6 (6.3)	-28.9 (4.7)	11.0 (9.4)	8.1 (5.0)	-17.3 (3.5)
DKK	-50.8 (47.9)	-62.5 (16.7)	-38.6 (4.4)	-30.1 (30.2)	-45.2 (10.9)	-39.4 (4.5)
CHF	-14.7 (14.3)	-28.1 (14.1)	-10.0 (3.9)	-12.8 (12.8)	-40.5 (10.2)	-14.4 (3.7)
SEK	-21.7 (21.0)	-20.5 (10.6)	-13.0 (3.5)	-7.2 (7.9)	0.4 (8.6)	-8.0 (2.5)
NOK	-25.8 (25.3)	-32.0 (17.2)	-11.7 (3.5)	-14.8 (12.3)	-19.0 (9.3)	-8.5 (2.9)
ILS	0.0 (0.0)	-62.4 (65.3)	-61.1 (21.5)	0.0 (0.0)	-57.0 (45.3)	-57.4 (11.0)
HKD	-4.6 (10.0)	-15.2 (10.0)	-17.8 (7.7)	-15.3 (18.0)	-5.3 (12.3)	-16.8 (6.9)
SGD	-6.9 (5.1)	-5.5 (2.8)	-1.5 (2.1)	-19.6 (15.9)	-29.8 (13.0)	-17.1 (4.8)
THB	-12.1 (13.2)	-17.0 (8.4)	-3.3 (1.7)	-9.8 (12.6)	-71.2 (41.9)	-16.8 (11.9)
MYR	-57.4 (51.7)	-74.9 (30.6)	-74.7 (19.5)	-95.2 (61.3)	-115.8 (29.9)	-77.1 (22.3)
KRW	-201.8 (115.3)	-120.3 (45.3)	-76.5 (32.8)	-114.2 (74.0)	-127.0 (43.0)	-86.0 (19.2)
RUB	62.0 (321.6)	-68.3 (33.6)	-55.4 (13.8)	117.8 (252.8)	-81.6 (36.3)	-116.3 (10.6)
ZAR	9.5 (27.9)	13.7 (15.2)	25.3 (39.3)	4.7 (26.8)	-20.8 (31.9)	-12.5 (26.5)
TRY	-120.6 (71.4)	-146.6 (62.1)	271.5 (331.7)	-115.0 (63.4)	-165.3 (39.8)	34.6 (99.8)
AED	-3.7 (20.7)	-107.2 (39.7)	-26.9 (8.4)	-6.5 (17.7)	-94.6 (33.3)	-37.0 (12.9)
SAR	-1.2 (6.5)	-50.5 (12.4)	-44.4 (11.2)	-1.6 (8.8)	-64.0 (17.1)	-64.2 (18.3)
CLP	-26.5 (60.4)	-61.8 (46.9)	-47.6 (20.2)	-12.0 (22.7)	-60.2 (16.7)	-40.6 (20.4)
MXN	-29.7 (38.5)	-103.7 (30.0)	-19.0 (19.1)	-53.0 (22.4)	-90.5 (31.5)	-26.8 (19.4)
CNY	-24.4 (67.8)	-315.3 (149.6)	-41.0 (9.4)	-36.7 (96.0)	-321.4 (125.3)	-45.0 (15.5)
QAR	0.0 (0.0)	-24.4 (34.2)	25.7 (0.9)	0.0 (0.0)	-36.6 (51.4)	-18.9 (0.1)

Table 2: Statistics of the XCCY in mean and standard deviations (in parentheses).

In other words, the USD was less special compared to these currencies.

The Chilean peso (CLP) and the Mexican peso (MXN) also appear positive, but it should be noted that their bases are quoted on the dollar leg, as is conventional in the market for these currencies, rather than the non-dollar leg, which is usually the case for other currencies. Thus, even though they seem positive, they are more aligned with negative basis currencies when quoted on the non-dollar leg, as is common in the swap market. Therefore, they do not belong to the class of well-known positive basis currencies, i.e., AUD and NZD.

We compute daily changes in the basis by the difference in the successive daily XCCY. Table 3 reports a set of summary statistics for the daily changes in the XCCY with 1- and 5-year maturities for the full sample between January 2000 and December 2020. We report the mean (scaled by 100), standard deviation, skewness, and excess kurtosis. From the summary statistics, we notice that the daily changes on a 1-year basis have a zero mean and a relatively high standard deviation. For developed markets, the standard deviation ranges between 0.99

Stats	1-Year Maturity XCCY				5-Year Maturity XCCY			
	Mean	Sdev	Skew	Kurt	Mean	Sdev	Skew	Ex-Kurt
EUR	-0.10	1.91	-1.54	125.69	-0.16	0.89	-0.26	34.51
JPY	-0.29	1.63	-0.21	58.81	-0.32	1.06	0.03	46.41
GBP	0.07	1.66	-3.22	165.13	0.10	0.75	0.30	36.13
AUD	-0.14	1.19	0.92	222.55	0.03	1.08	-0.67	131.25
NZD	-0.08	0.99	0.07	38.60	0.08	0.85	0.30	37.11
CAD	-0.48	1.10	-2.07	66.13	-0.34	0.94	-22.72	1254.79
DKK	-0.47	2.22	-3.03	128.74	-0.49	1.23	-2.61	99.71
CHF	-0.15	1.68	1.08	90.50	-0.13	0.89	0.21	48.07
SEK	-0.14	1.19	-1.64	45.00	-0.06	0.62	2.42	62.27
NOK	-0.03	1.39	-6.43	395.53	-0.02	0.64	-0.33	83.54
ILS	-0.78	3.76	-26.67	1543.70	-1.03	2.10	-17.19	837.18
HKD	-0.48	1.59	0.80	29.34	-0.62	1.28	-0.91	28.00
SGD	-0.03	1.22	-0.62	126.40	-0.65	1.26	-3.87	109.29
THB	-0.07	1.40	-1.42	89.46	-0.08	2.37	-0.89	53.51
MYR	-0.90	4.91	4.59	164.20	-0.94	3.33	-0.19	26.49
KRW	-0.78	7.61	-1.93	62.05	-1.12	6.54	-1.41	28.17
RUB	-1.28	24.26	1.97	201.90	-1.78	19.97	-3.93	253.04
ZAR	1.67	3.31	0.24	18.53	0.53	2.48	0.21	30.44
TRY	-2.37	16.09	3.09	121.87	-0.06	10.24	0.44	90.87
AED	-0.38	3.29	-7.72	298.46	-0.53	3.29	-0.82	41.35
SAR	-1.01	3.64	0.33	48.64	-1.30	4.47	0.43	32.43
CLP	-0.66	4.92	-1.72	86.31	-0.32	2.29	-2.05	58.54
MXN	-0.00	4.55	1.01	93.41	0.00	2.67	-1.67	167.16
CNY	-0.56	7.71	-3.91	100.65	-0.58	10.85	-1.07	132.85
QAR	0.41	1.84	-8.11	405.74	-0.29	4.35	-2.85	166.37

Table 3: Summary statistics of daily changes in XCCY (Jan 2000–December 2020).

(New Zealand dollar - NZD) and 3.78 (Israeli shekel - ILS), while for emerging markets, it ranges between 1.40 (Thai baht - THB) and 24.26 (Russian ruble - RUB). The highest standard deviations are observed for Russia (RUB) and Turkey (TRY).

The skewness of the daily changes on a 1-year basis ranges from -26.67 (Israeli shekel - ILS) to 3.09 (Turkish lira - TRY), with the majority being negative. The excess kurtosis varies between 18.63 (South African rand - ZAR) and 1523.39 (Israeli shekel - ILS). Similarly, the daily changes in the 5-year XCCY also show a zero mean and a relatively high standard deviation. For developed markets, the standard deviation ranges between 0.62 (Swedish krona - SEK) and 19.97 (Russian ruble - RUB). The second most volatile basis after the Russian ruble (RUB) is the Chinese yuan (CNY). The skewness of the daily changes in the 5-year XCCY ranges from -22.72 (Canadian dollar - CAD) to 2.42 (Swedish krona - SEK), with the majority being negative. The excess kurtosis varies between 26.49 (Malaysian ringgit - MYR) and 1254.79 (Canadian dollar - CAD). Overall, the daily changes in both the 1- and 5-year

XCCY show a leptokurtic behaviour, characterised by high peaks and fat tails compared to a normal distribution.

5. Empirical Findings and Discussion

Before turning to our proposed network-VAR estimation, it is worth noting that pairwise correlations among cross-currency basis changes already suggest substantial co-movement across currencies, particularly during episodes of global stress such as the GFC and COVID-19. However, such simple pairwise correlations are unable to distinguish direct from indirect linkages and cannot account for the multilateral structure of funding relationships. For this reason, we employ a graphical VAR framework that conditions on all other currencies simultaneously, allowing us to identify the network of direct contemporaneous and dynamic dependencies rather than aggregate co-movements. In contrast to unconditional correlations, the graphical VAR reveals both the direction and structure of interdependence of the basis spreads, providing a clearer view of how funding shocks propagate through the global basis network.

Accordingly, we apply our proposed estimation methodology to analyse the interconnect-
edness among the G-25 cross-currency basis swap spreads using a rolling window of one year (approximately 240 trading days). This rolling window approach captures the annual dependencies among basis spreads, with monthly increments starting from February 2000 to January 2001 and ending from July 2019 to December 2020. This results in a total of 234 rolling windows.

To understand the dynamics of basis interconnectedness, we use network density and the turbulence index. Network density measures the proportion of actual connections to the total possible connections within the network, indicating the strength of mutual influence among the basis spreads. The turbulence index, adapted from [Ahelegbey and Giudici \(2022\)](#) and [Kritzman and Li \(2010\)](#), assesses volatility and structural changes.

The optimal lag for the VAR model is determined using the Bayesian Information Criterion (BIC) for different lag orders ($p \in \{1, \dots, 7\}$). Table 4 presents the BIC scores, with the optimal lag order indicated in bold. Based on these results, we select a lag order of $p = 1$.

p	1	2	3	4	5	6	7
BIC (1Y)	48.15	48.27	48.58	48.96	49.30	49.65	50.10
BIC (5Y)	35.79	36.25	36.75	37.30	37.89	38.51	39.11

Table 4: The BIC for lag order selection. Boldface values indicate the minimum BIC

5.1. Network Density and Financial Turbulence

We address our first research question (RQ-1): Do cross-currency basis swap spreads reflect systemic contagion or merely isolated fluctuations?

Network density and the turbulence index provide insights into this question. Network density, representing the proportion of observed connections relative to possible connections, reveals the degree of interconnectedness among basis spreads. The turbulence index, which measures the extent of unusual changes in asset returns and interactions, signals periods of market stress.

Figure 3 shows the time series of network density and the turbulence index for both 1-year and 5-year maturities. Both measures show significant spikes during key crises: the Global Financial Crisis (GFC), the European Sovereign Crisis (ESC), the oil price shock of 2014-2016, the tax reform impact of late 2017 to mid-2018, and the COVID-19 pandemic. The turbulence index reached its peak during the GFC, with notable but lower peaks during subsequent crises, including the COVID-19 pandemic.

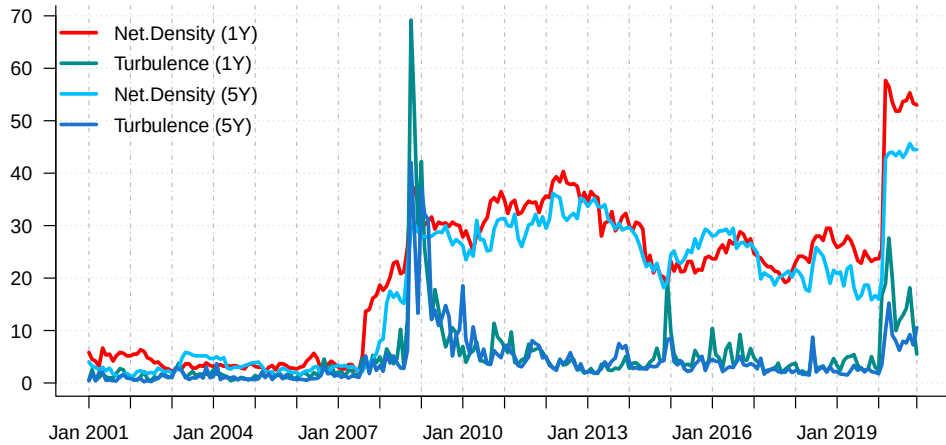


Figure 3: Network Density and Turbulence Index among XCCY with 1 & 5-year maturities.

The analysis of network density and the turbulence index over the rolling windows reveals significant variations in the interconnectedness and stability of the G-25 XCCY network over

time. Key findings include:

1. Pre-Crisis Period (2000–2006): During this period, the network density was relatively low, indicating limited interconnectedness among the basis spreads. The turbulence index was also low, suggesting a stable market environment.
2. Global Financial Crisis (2007–2009): The network density increased significantly, reflecting heightened interconnectedness among the basis spreads. The turbulence index spiked, indicating substantial market stress and structural shifts within the network.
3. Post-Crisis Period (2010–2013): Although the network density decreased slightly from its crisis peak, it remained elevated compared to the pre-crisis period. The turbulence index showed periodic spikes, reflecting ongoing market uncertainties, particularly during the Eurozone crisis.
4. Recent Period (2014–2020): The network density showed a gradual increase, indicating growing interconnectedness among the basis spreads. The turbulence index exhibited occasional spikes, notably during the COVID-19 pandemic, reflecting heightened market volatility and stress.

The severity of turmoil in the foreign exchange currency derivatives market, as tracked by the turbulence index, recorded its historical highest spike during the 2007–2009 global financial crisis, with moderate spikes during the oil crisis and the recent COVID-19 pandemic. Although the severity of the turmoil in the currency derivatives market was much higher during the GFC compared to the COVID-19 period, the degree of interconnectedness during the latter was much greater than during the former. Therefore, if the market turbulence seen during the COVID-19 pandemic had not been taken seriously and allowed to continue, the devastating consequences in the foreign derivatives market could have been much larger than ever seen, even surpassing the worst of the global financial crisis.

In the interest of addressing whether the movements in the basis swap spreads provide evidence of contagion, Figure 4 shows a plot comparing the average cross-currency basis swap spread against the Net-Density and Turbulence index. The figure shows that the more negative the average cross-currency basis swap spreads, the higher the significant rise in market network density and turbulence. This suggests strong evidence of contagion in all the

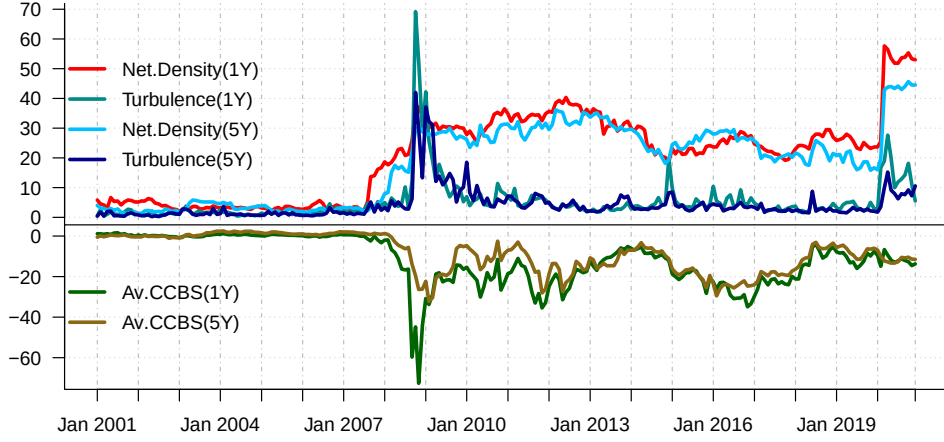


Figure 4: Network Density, Turbulence Index and Average Cross-Currency Basis Swap Spread of G-10. This figure plots the monthly average cross-currency basis swap spreads, the degree of interconnectedness, and the turbulence index between January 2001 and December 2020.

x_{t+h}	-4	-3	-2	-1	0	1	2	3	4
Cross Correlation of Δ Net.Density (1Y) with									
Δ Av.XCCY (1Y)	0.023	-0.020	0.065	0.066	-0.312	-0.008	0.030	0.053	0.033
Cross Correlation of Δ Turbulence (1Y) with									
Δ Av.XCCY (1Y)	-0.011	0.129	-0.067	0.131	-0.315	-0.302	0.105	0.137	0.024
Cross Correlation of Δ Turbulence (1Y) with									
Δ Net.Density (1Y)	0.036	-0.200	-0.054	0.062	0.318	0.048	-0.090	0.027	-0.035
Cross Correlation of Δ Net.Density (5Y) with									
Δ Av.XCCY (5Y)	-0.079	-0.058	-0.017	0.125	-0.109	-0.189	-0.150	-0.005	0.014
Cross Correlation of Δ Turbulence (5Y) with									
Δ Av.XCCY (5Y)	0.021	0.025	0.066	-0.068	0.120	-0.425	0.124	-0.084	0.035
Cross Correlation of Δ Turbulence (5Y) with									
Δ Net.Density (5Y)	-0.079	-0.007	-0.086	0.016	0.254	0.012	-0.046	0.013	0.011

Table 5: Cross Correlation of an average daily basis, network density, and turbulence index for both maturities. The result shows the correlation of x_{t+h} (the row variable) with y_t (the column variable). Boldface values indicate the significant correlations.

identified crisis periods, and did not occur at the same magnitude during the more tranquil times in the foreign exchange derivatives market. Through a cross-correlation analysis, Table 5 confirms the negative relationship between the movements in the cross-currency basis swap spreads and the market network density and turbulence. It can be seen in the table, that the most significant cross-correlation of the average cross-currency basis swap spreads (Av.XCCY) with the network density and turbulence occurs at lag 0 for the 1-year maturity and lag 1

for the 5-year maturity. The network density and turbulence index for both maturities, however, exhibit a positive contemporaneous relationship. Thus, the evidence indicates that periods of very high negative deviations in CIP are often associated with a high degree of interconnectedness and turbulence in foreign exchange derivatives markets.

Based on the similarity in the behaviours of the network density and turbulence index, two stylised facts can emerge: 1) During the pandemic, the behaviour of cross-currency basis swap spreads in the foreign exchange derivatives markets is in several ways more similar to the GFC than it is to other periods of stress. Just like during the GFC, it appears that the foreign exchange derivative market is in a transition to a quieter period where stress is diminished in the market, dollar funding pressure is lowered and deviations from CIP are less acute; and 2) Based on the behavior of the network density and turbulence index during the GFC, the subsequent pattern in the foreign exchange derivatives market after the recent pandemic could be one where there is diminished unusualness, turbulence is quietened, and deviations from CIP become less pronounced, i.e. the basis becomes much less negative (tighter), especially given the bazooka of liquidity that the Federal Reserve has embarked upon, weakening implications this has for the US dollar and the enhancement of cross border flows. [Avdjiev et al. \(2019\)](#) documented a well-known triangular relationship, demonstrating that a weaker dollar correlates with increased banking flow, or the supply of US dollars in the foreign exchange derivatives markets, and a tighter cross-currency basis swap spread.

5.2. Network Topology and Centrality

We now turn our attention to the second research question: (RQ-2) In the event of contagion, which basis swap spreads are central to spillover propagation in the foreign exchange derivatives market?

To answer the question above, we will first describe the dynamic interconnectedness in the covered interest parity, monitoring the topological structure of the G-25 cross-currency basis swap spreads, and assessing the most critical (or central) basis for the foreign exchange derivatives market. We briefly present the metric for the network topology's construction and some standard network centrality measures used in this study.

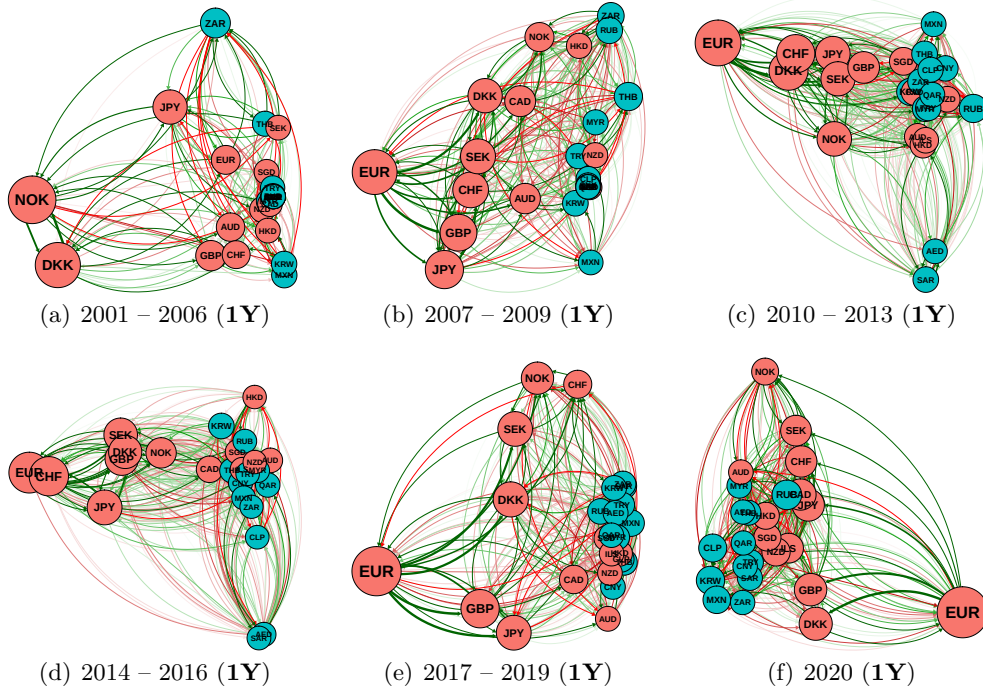


Figure 5: Networks of XCCY with **1-year** maturity. Red links denote negative effects and green links denote positive interactions. Red (blue) nodes are developed (emerging) market currencies.

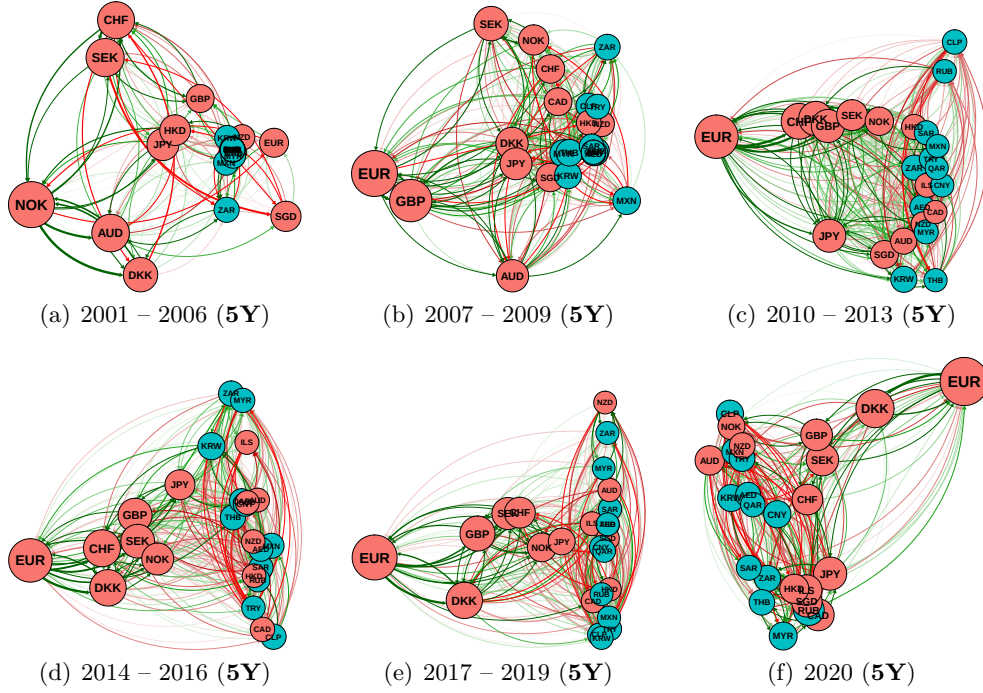


Figure 6: Networks of XCCY with **5-year** maturity. Red links denote negative effects and green links denote positive interactions. Red (blue) nodes are developed (emerging) market currencies.

In network analysis, node centrality captures the relative prominence or influence of a node within the system. Several metrics quantify this importance, including in-degree and out-degree (which count the number of incoming and outgoing connections, respectively), as well as authority and hub scores. Conceptually, a node’s authority score rises when it is pointed to by highly influential hubs, whereas a hub’s score increases when it connects to strong authorities. Mathematically, these measures can be derived from the eigen-decomposition of the adjacency matrices (AA') and $(A'A)$, with the magnitudes of the principal eigenvectors providing the respective authority and hub centralities. In financial networks, nodes with high in-degree or authority values can be interpreted as *targets of influence*, while nodes with high out-degree or hub scores act as *sources of influence* that transmit shocks or information to others (see [Abdelsalam et al., 2024](#) for related definitions of centrality metrics).

We report in Figures 5 and 6 the topological structure interconnectedness of 1-year and 5-year maturity basis over six time periods: (2000–2006), (2007–2009), (2010–2013), (2014–2016), (2017–2019), and (2020). Network links are colour-coded, with red links indicating a negative effect and green links meaning positively weighted edges. The nodes are also colour-coded, such that red nodes represent developed market currencies and blue nodes represent emerging markets. The size of the nodes is proportional to their hub scores.

The network topology shows evidence of clustering behaviour among the currencies. More importantly, the red-colored nodes (the developed market currencies) seem to move together due to similarities in underlying basis conditions. Likewise, the blue-color nodes (the emerging market currencies) also tend to move together due to similarities in underlying conditions. Except for the 2001–2006 network, which has NOK (Norwegian krone) and the DKK (Danish krone) as the largest-sized nodes (i.e., the most influential), the rest of the sub-period networks show EUR (the Euro) as the largest-sized node (i.e., the most influential).

Table 6 shows a summary of the centrality rankings of the most important currencies over the sub-periods and a full sample based on the median statistics of the hub and authority scores. This is done to find the most important cross-currency basis swap spread. The table shows the top three most influential currencies for the sub-periods, and the full sample ranking from “core” (1st to 5th), through “semi-periphery” (6th to 15th), to “periphery” (16th

		1-Year Maturity XCCY		5-Year Maturity XCCY	
	Rank	Hub	Auth	Hub	Auth
Sub-Periods: Top Three Most Influential Currencies					
2001 – 2006	1	ZAR (0.423)	JPY (0.267)	SEK (0.243)	NOK (0.243)
	2	HKD (0.291)	KRW (0.244)	HKD (0.219)	AUD (0.200)
	3	NOK (0.264)	ZAR (0.239)	AUD (0.176)	ZAR (0.181)
2007 – 2009	1	JPY (0.291)	JPY (0.274)	JPY (0.279)	JPY (0.277)
	2	EUR (0.286)	DKK (0.253)	CAD (0.242)	DKK (0.269)
	3	DKK (0.251)	MXN (0.247)	GBP (0.238)	SGD (0.258)
2010 – 2013	1	EUR (0.269)	SEK (0.272)	EUR (0.277)	EUR (0.297)
	2	CHF (0.256)	EUR (0.255)	SEK (0.266)	JPY (0.254)
	3	SEK (0.255)	JPY (0.251)	JPY (0.225)	CHF (0.237)
2014 – 2016	1	CHF (0.282)	JPY (0.273)	EUR (0.252)	KRW (0.251)
	2	JPY (0.249)	CLP (0.248)	ZAR (0.249)	EUR (0.249)
	3	EUR (0.232)	SEK (0.218)	CNY (0.245)	RUB (0.238)
2017 – 2019	1	EUR (0.304)	EUR (0.306)	EUR (0.293)	DKK (0.277)
	2	DKK (0.299)	CHF (0.295)	DKK (0.268)	GBP (0.257)
	3	CHF (0.260)	DKK (0.282)	GBP (0.264)	EUR (0.243)
2020	1	KRW (0.247)	JPY (0.262)	KRW (0.315)	ZAR (0.267)
	2	EUR (0.245)	KRW (0.251)	JPY (0.273)	TRY (0.253)
	3	GBP (0.239)	NOK (0.248)	AUD (0.270)	JPY (0.252)
Full-Sample: All G-25 Currencies					
Core	1	EUR (0.253)	JPY (0.260)	EUR (0.229)	JPY (0.223)
	2	DKK (0.243)	NOK (0.238)	CHF (0.219)	EUR (0.221)
	3	CHF (0.239)	SEK (0.227)	JPY (0.211)	CHF (0.218)
	4	JPY (0.227)	DKK (0.220)	DKK (0.197)	KRW (0.216)
	5	ZAR (0.220)	EUR (0.219)	SEK (0.192)	GBP (0.212)
Semi-Periphery	6	NOK (0.216)	KRW (0.216)	KRW (0.190)	ZAR (0.203)
	7	SEK (0.210)	GBP (0.213)	ZAR (0.189)	DKK (0.193)
	8	GBP (0.207)	CHF (0.204)	MYR (0.184)	SEK (0.171)
	9	HKD (0.192)	ZAR (0.188)	GBP (0.171)	NOK (0.169)
	10	MXN (0.187)	HKD (0.176)	HKD (0.169)	HKD (0.157)
	11	KRW (0.178)	MXN (0.167)	AUD (0.164)	MYR (0.157)
	12	RUB (0.169)	RUB (0.155)	RUB (0.154)	RUB (0.155)
	13	MYR (0.154)	AUD (0.151)	THB (0.148)	THB (0.151)
	14	THB (0.128)	CAD (0.148)	MXN (0.130)	MXN (0.150)
	15	CAD (0.124)	MYR (0.144)	NOK (0.121)	CLP (0.145)
Periphery	16	AUD (0.123)	THB (0.126)	TRY (0.121)	SGD (0.142)
	17	TRY (0.114)	NZD (0.117)	CLP (0.121)	AUD (0.137)
	18	SGD (0.104)	TRY (0.113)	SGD (0.120)	TRY (0.131)
	19	SAR (0.103)	CNY (0.096)	SAR (0.115)	AED (0.114)
	20	NZD (0.101)	SGD (0.078)	NZD (0.093)	CAD (0.106)
	21	CLP (0.071)	SAR (0.077)	CAD (0.092)	NZD (0.099)
	22	CNY (0.071)	AED (0.051)	AED (0.084)	SAR (0.090)
	23	AED (0.049)	CLP (0.044)	CNY (0.075)	CNY (0.067)
	24	ILS (0.000)	ILS (0.000)	ILS (0.000)	ILS (0.000)
	25	QAR (0.000)	QAR (0.000)	QAR (0.000)	QAR (0.000)

Table 6: Centrality ranking 1-year and 5-year swaps of according to hub and authority score.

to 25th). The result shows that central to the network of interconnectedness for spillovers and propagation of contagion is the Euro, with the highest hub score for both 1-year and 5-year maturity bases. The Japanese yen (JPY) is the most central in terms of authority score for both 1-year and 5-year maturity bases.

Over the full sample, the most influential (in terms of hub score ranking) currency for spillovers and propagation is the Euro. It is usually within the top three highest hub-score currencies over the various sub-periods. The next most influential 1-year maturity basis is the Danish krone (DKK), followed by the Swiss franc (CHF), the Japanese yen (JPY), and the South African rand (ZAR). The next most influential 5-year maturity basis after the Euro, according to the full sample statistics, is the Swiss franc (CHF), followed by the Japanese yen (JPY), the Danish krone (DKK), and the Swedish krona (SEK). A surprising finding from the hub ranking is the South African rand - ranked 5th for the 1-year maturity, and 7th highest hub score for the 5-year maturity.

The summary ranking of the authority scores for the 1-year maturity shows that the top 5 bases are the Japanese yen (JPY), the Norwegian krone (NOK), the Swedish krona (SEK), the Danish krone (DKK), and the Euro. The top 5 authority scores for the 5-year maturity are the Japanese yen (JPY), the Euro, the Swiss franc (CHF), the Korean won (KRW), and the British pound (GBP). Another surprising finding from the authority ranking is the Korean won - ranked 6th for the 1-year maturity, and 4th highest authority score for the 5-year maturity.

Overall, the result shows that central to the CIP interconnectedness is the finding that most of the spillovers propagate from the Euro, the Swiss franc, and the Danish krone bases to the other major currencies. Interestingly, the Swiss franc and the Danish krone are pegged closely to the Euro, ultimately making the most influential basis Euro-centred. This suggests that shocks that impair the functioning of the global cross-currency basis swap markets first manifest in the Euro currency basis before spreading to others, making the Euro cross-currency basis swap have the most influence on the variability of the bases of other major currencies. One plausible reason, we believe, is that the euro-dollar pair, and by extension its associated basis, being the most liquid in the world, partly explains this phenomenon. The

euro basis vis-à-vis the dollar is the most actively traded cross-currency basis swap spread in the global derivatives market.

While earlier studies have documented that crises amplify co-movements in other foreign-exchange markets, the mechanism we study is distinct. For instance, [Karnaukh et al. \(2015\)](#) show that FX spot-market liquidity measures co-move more strongly under funding stress and high volatility, and [Kloks et al. \(2023\)](#) report similar liquidity commonality in FX swaps. However, these papers focus on *market-liquidity co-movement*—correlated bid-ask spreads and transaction costs—whereas our analysis concerns *balance-sheet co-movement* in the cross-currency basis, a pricing measure of funding constraints and intermediary capacity. The cross-currency basis does not simply reflect transient order-flow liquidity; it captures the shadow price of dealer balance-sheet usage that equilibrates global dollar funding. Hence, our finding that basis spreads move together in crises complements, rather than replicates, the prior literature: it identifies a structural balance-sheet channel through which systemic stress synchronizes deviations from covered interest parity across currencies.⁴

5.3. Implication For Market Participants

Here we address our third and final research question: (RQ-3) What are the implications for financial market participants and policymakers?

The finding shows that shocks in the global cross-currency basis swap markets first show up in the euro currency swap markets before spreading to others. This means that the euro cross-currency basis swap has the most impact on how the bases of other major currencies change. One plausible reason, we believe, is that the euro-dollar pair, and by extension its associated basis, being the most liquid in the world, partly explains this phenomenon. The euro basis vis-à-vis the dollar is the most actively traded cross-currency basis swap spread in the global derivatives market. One implication of our finding is that financial market participants need to track events on the euro basis (together with the bases of other currencies pegged/linked closely to it) much more carefully than before. This is because the euro basis is central to

⁴Unlike FX spot or swap liquidity co-movements, which arise from temporary microstructure frictions, basis co-movements stem from balance-sheet constraints that persist across maturities and regulatory regimes. See [He et al. \(2017\)](#) for intermediary-asset-pricing evidence consistent with this mechanism.

contagion in the cross-currency swap markets, and sets the stage for the dynamics observed on the other basis swap spreads. Therefore, empirical and theoretical modelling of cross-currency swap markets cannot exclude the euro basis. Apart from providing a benchmark in the market, the euro basis is the most liquid basis swap spread, accounting for a large number of trading volumes, and has been shown to have the most tactical influence on the other basis swap spreads.

For agents with exposure to the foreign exchange derivatives markets, there are several financial market implications of the stylised facts derived from the above findings. First, for dollar entities that look to enter new hedges against exchange rate risk for their foreign currency exposure, it is less expensive and more cost-effective to do those hedges before the cross-currency basis swap spread associated with their intended currencies of exposure becomes tighter. This will help to lower basis risk and increase the marked-to-market gains from the basis going up (then tightening). One possible arbitrage trade that could make money when the basis is more negative (but before it gets tighter or less negative) would be to borrow in the currency that has a higher interest rate (the US dollar) and lend in the currency that has a lower interest rate and a negative basis (the Euro). This way, one would be well protected against foreign exchange risk. This is the opposite of the traditional carry trade. Second, borrowers of these foreign currencies, who first borrow US dollars before entering a foreign exchange derivatives contract to swap USD for their preferred foreign currency, may find it more cost-effective to do so when the cross-currency basis swap spread is wider, rather than when it becomes tighter, as the stylised fact suggests. As a result, these foreign currencies can be raised more cheaply.

For currencies with a positive basis, the advantage or profitability comes after the basis has become tighter. For entities with these currencies looking to enter into new hedges against exchange rate risk for their dollar exposure, it is less expensive and more cost-effective to implement those hedges after the basis has become tighter or more positive. In this case, the profitable trade is to borrow in the high-interest-rate currency (the one with a positive basis, i.e. AUD) and lend in a lower-interest-rate currency (i.e. US dollar), with exposure to foreign exchange risk well hedged. Borrowers of the US dollar can also borrow more cheaply by first

borrowing in the currency of a positive basis and swapping to US dollars while the basis of this currency is still mostly positive.

Given potentially severe and long-lasting effects towards financial stability, conditional on the network effects and degree of transmission of our findings, sustained violations of the CIP should be the focus of coordinated and pro-active intervention by the leading central banks of the world. Since, as our results show, there is a special role of the euro currency in propagating shocks throughout the financial eco-system, the European Central Bank should re-assess its role of using foreign-exchange rates as direct instruments to maintain its newly found objective of financial stability. Furthermore, given the degree of financial interconnectedness and monetary interdependence, central banks and international monetary institutions should work together in order to solve the issues leading to substantial and persistent deviations from CIP like increased counterparty risk and shortages in liquidity (Atay, 2006; Baba et al., 2008; Cunado et al., 2024). One solution might be coordinated action to provide short-term liquidity should the CIP diverge from its pre-crisis level.

5.4. Practicality of Our Results' Implications

Our results indicate that shocks in global cross-currency basis markets typically surface first in the euro basis before propagating more broadly. A plausible reason is depth and liquidity: the EUR–USD swap market is the most actively traded segment, and balance-sheet linkages within the European banking system tie closely to the euro leg. Practically, this means the euro basis (and closely linked currencies such as DKK and CHF) is an especially informative bellwether for impending shifts in broader basis conditions. Empirical and theoretical models that omit the euro leg risk missing the most influential channel of transmission.

For market participants with FX-derivatives exposures, three implications follow:

1. **Timing of hedge implementation.** When basis spreads are wide and volatile, hedge costs and mark-to-market variability rise. Entities that anticipate tightening (a less negative basis on USD legs, or a more positive basis on non-USD legs) can reduce expected hedge costs and basis risk by locking terms before conditions tighten, subject to collateral, margin, and funding constraints.

2. **Funding route selection.** For a given currency exposure, the relative cost of direct funding versus synthetic funding via cross-currency swaps is state dependent. Monitoring the euro basis and its close satellites improves forecasts of these relative costs and helps identify when the swap route is likely to be more economical. (Actual realizations depend on term, CSA terms, haircuts, and balance-sheet charges.)
3. **Heterogeneity across currencies.** Currencies with persistently positive bases (e.g., AUD/NZD historically) and USD-pegged currencies behave differently. For the former, synthetic USD funding can at times be cheaper than cash markets; for tightly pegged legs, lower idiosyncratic demand and weaker bilateral linkages imply muted volatility and limited spillovers. Risk management, therefore, should distinguish between (i) bellwether transmitters (EUR/CHF/DKK), (ii) receivers, and (iii) insulated pegged legs.

6. Broader Economic Interpretation of the Network Dynamics

The network results reveal an economic structure consistent with the mechanisms that underpin global dollar funding markets. Periods of elevated interconnectedness and turbulence coincide with global stress episodes such as the GFC and COVID-19, reflecting systemic tightening in dollar liquidity and balance-sheet constraints among international dealers. When intermediation capacity becomes scarce, funding costs rise simultaneously across currencies, producing the high co-movement and dense networks observed in the data. These episodes therefore represent global, supply-side shocks to funding rather than a simple increase in correlated demand for U.S. dollars.

The bilateral connections that dominate the network—particularly those radiating from the euro and its closely linked currencies such as the Danish krone and the Swiss franc—capture deep institutional and balance-sheet linkages in the European banking system. These currencies act as hubs through which funding shocks are transmitted to other markets. Their prominence is consistent with the concentration of dealer activity and collateral reuse within the euro-dollar swap ecosystem, where constraints on European intermediaries can rapidly propagate to the rest of the world.

At the same time, currencies with credible dollar pegs or abundant reserve buffers, such as the Hong Kong dollar or Saudi riyal, remain relatively insulated. Their limited net demand for dollar funding dampens their exposure to network spillovers, explaining their peripheral position during stress periods. This asymmetry between central transmitters and peripheral absorbers highlights how institutional arrangements and policy regimes shape the topology of the global funding network.

The maturity dimension of the basis further supports this interpretation. Shorter-tenor bases respond more sharply to liquidity and margin shocks, while longer-tenor bases are driven by structural balance-sheet costs and exhibit weaker spillovers. The pattern of declining volatility and contagion with maturity reflects how short-term funding stress is absorbed and gradually dissipates along the tenor curve.

Taken together, the network indicators provide an interpretable map of global dollar funding conditions. Rising density and turbulence correspond to periods when balance-sheet constraints and collateral scarcity intensify, while declining interconnectedness signals normalization and a return of intermediation capacity. In this sense, network metrics serve as early-warning indicators of systemic stress—observable manifestations of the underlying economic forces that govern international liquidity. Central-bank swap lines, coordinated liquidity facilities, and margin-relief measures can therefore be viewed as instruments that directly relax these constraints, compressing network density and limiting contagion across funding markets.

7. Conclusion

This paper establishes the existence, nature, and extent of interconnectedness among the cross-currency basis swap spreads of the major 25 world currencies (G-25). We examine the strength of interconnections and the degree of contagion using financial network analysis, which provides a novel method to study linkages in the currency derivatives market and assess how these networks have evolved over different periods.

Our findings strongly suggest that changes in deviations from covered interest parity for currencies are linked during times of peace, times of crisis, and times after a crisis, when

changes are more noticeable. During crisis periods, this interconnectedness is notably stronger and more persistent, suggesting potential mark-to-market losses for investors with long basis exposures. The systematic increase in interconnectedness among cross-currency basis swap spreads creates larger arbitrage opportunities across currency and fixed-income markets, allowing investors with US dollars to obtain foreign currencies more cost-effectively by issuing in US dollars and swapping to foreign currencies via cross-currency swaps.

We find that the derivatives market experienced its highest levels of turmoil and interconnectedness spikes during the global financial crisis and the COVID-19 pandemic compared to the eurozone crisis, suggesting stronger contagion in cross-currency basis swap markets during these major financial disruptions. Notably, spillovers predominantly propagate from the Euro, Swiss franc, and Danish krone bases to other major currencies, highlighting the Euro’s central role in driving contagion among bases.

To our knowledge, this is the first study to combine the cross-currency basis swap spreads of 25 currencies, both across maturities and with a Bayesian graphical SVAR that recovers a directed spillover network for the cross-currency basis swap spreads.

We provide a theory-plus-evidence account of the cross-currency basis swap spreads with four predictions: peg immunity, a maturity gradient (lower volatility/spillovers at longer tenors), strong co-movements in crisis-time, and transmission governed by centrality, which we verify on 25 currencies using a Bayesian graphical SVAR that recovers a sparse, directed spillover network. We demonstrate that basis contagion is Euro-centred (with CHF/DKK closely tied), and USD-pegged legs act as shock absorbers. Systemic episodes compress the market toward a single factor, providing a micro-founded map of how dollar scarcity propagates across funding markets. Our contribution is both conceptual and practical: it explains when and where CIP breaks, identifies who transmits and who absorbs stress, and provides actionable guidance for designing swap lines, liquidity backstops, and basis-risk management.

It is important to note that this study does not have a goal to empirically directly measure and differentiate the specific drivers or transmission mechanisms of interconnectedness or contagion across the cross-currency basis swap spreads analysed here. As noted by [Forbes and Rigobon \(2002\)](#), comprehensively measuring these transmission mechanisms is challenging,

and we suggest this as an area for future research.

Acknowledgements

This paper builds on and extensively expands an earlier working paper entitled *Interconnected Deviations from Covered Interest Parity* (Ahelegbeya and Ibhagui, 2022), which circulated as DEM Working Paper No. 191 at the University of Pavia. The authors thank the editors for noting this connection.

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Disclosure of interest

The authors declare no competing interests or conflicts of interest.

Funding

No funding or financial support was received for the conduct of this research or the preparation of this article.

Data availability statement

The data that support the findings of this study were obtained from Bloomberg, a private, subscription-based data provider.

These data are not publicly available due to licensing restrictions but may be accessed on a confidential basis from the corresponding author, subject to the rules and conditions of Bloomberg, the data provider.

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A. Proofs of the Propositions

A.1. Setup and Standing Assumptions

Let n denote the number of currencies. In each period t , client net USD demand across currencies is $d_t = (d_{1,t}, \dots, d_{n,t})' \in \mathbb{R}^n$. Dealers clear these flows one-for-one, so equilibrium quantities are $q_t = d_t$. The per-period quadratic cost of absorbing positions is

$$C_t(q_t) = \frac{1}{2} q_t' (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') q_t$$

where $T = \text{diag}(\tau_1, \dots, \tau_n)$, $\Phi = (\phi_{ij})$ is symmetric with $\phi_{ij} \geq 0$, $\Lambda_t \geq 0$ is a scalar, and $\mathbf{1}$ is the $n \times 1$ vector of ones. Dealers are risk-neutral and competitive, so the CIP basis vector equals marginal cost evaluated at the clearing quantities:

$$b_t = \nabla_{q_t} C_t(q_t)|_{q_t=d_t} = (T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}') d_t \equiv K_t d_t, \quad (\text{A.1})$$

where $K_t := T + \Phi + \Lambda_t \mathbf{1}\mathbf{1}'$ is the cost kernel.

Assumption 1 (Positivity and Boundedness). *T is diagonal with $\tau_k \geq 0$, Φ is symmetric with $\phi_{ij} \geq 0$, and both T and Φ are time-invariant. There exist constants $0 < m \leq M < \infty$ such that $mI \preceq T + \Phi \preceq MI$ for all t . This ensures $T + \Phi$ is positive definite and well-conditioned.*

Assumption 2 (Demand Moments). *The demand vector satisfies $\mathbb{E}[d_t] = 0$ and $\Sigma_d := \text{Var}(d_t) \succ 0$ with finite variance. The global stress parameter Λ_t is independent of d_t and satisfies $\mathbb{E}[\Lambda_t^2] < \infty$.*

Assumption 3 (Maturity Heterogeneity). *Inventory costs τ_k increase with maturity, reflecting longer balance-sheet commitments for longer-tenor swaps. Demand is concentrated at the short end, i.e., for maturities $m_2 > m_1$, $\text{Var}(d^{(m_2)}) \leq \alpha \text{Var}(d^{(m_1)})$ with $0 < \alpha < 1$. Cross-currency linkages ϕ_{ij} attenuate at longer maturities, as dealers warehouse long-dated exposures more idiosyncratically. These conditions imply that both the variance and the spillover intensity of $b_t^{(m)}$ decline with maturity.*

Notation.. For $x \in \mathbb{R}^n$, x_k denotes the k -th component. For a matrix A , A_{ij} is its (i, j) element, $A_{k\bullet}$ its k -th row, and $A_{\bullet j}$ its j -th column. $\|\cdot\|$ denotes the operator norm.

A.2. Proof of Proposition 1

Proof. From (A.1), the basis vector satisfies

$$b_t = K_t d_t = (T + \Phi) d_t + \Lambda_t \mathbf{1} \mathbf{1}' d_t = (T + \Phi) d_t + \Lambda_t S_t \mathbf{1}, \quad S_t := \mathbf{1}' d_t$$

Fix two currencies $i \neq j$. Denote the idiosyncratic contributions as

$$a_{i,t} := ((T + \Phi) d_t)_i, \quad a_{j,t} := ((T + \Phi) d_t)_j$$

so that

$$b_{i,t} = a_{i,t} + \Lambda_t S_t, \quad b_{j,t} = a_{j,t} + \Lambda_t S_t$$

Introduce a scaling parameter $M > 0$ and define a scaled stress process

$$\Lambda_t(M) := M \Lambda_t(0)$$

where $\Lambda_t(0)$ is the baseline stress, independent of d_t . Let $\nu_0 := \mathbb{E}[\Lambda_t(0)^2]$ and $v_S := \text{Var}(S_t) > 0$ (Assumption A2). The covariance and variances are

$$\text{Cov}(b_{i,t}, b_{j,t}) = \text{Cov}(a_{i,t}, a_{j,t}) + \nu_0 M^2 v_S$$

$$\text{Var}(b_{i,t}) = \text{Var}(a_{i,t}) + \nu_0 M^2 v_S$$

$$\text{Var}(b_{j,t}) = \text{Var}(a_{j,t}) + \nu_0 M^2 v_S$$

Hence, the correlation is

$$\text{Corr}_M(b_{i,t}, b_{j,t}) = \frac{\text{Cov}(a_{i,t}, a_{j,t}) + \nu_0 M^2 v_S}{\sqrt{(\text{Var}(a_{i,t}) + \nu_0 M^2 v_S)(\text{Var}(a_{j,t}) + \nu_0 M^2 v_S)}}.$$

As $M \rightarrow \infty$, the $\nu_0 M^2 v_S$ term dominates numerator and denominator. Dividing numerator and denominator by $\nu_0 M^2 v_S$ and taking the limit gives

$$\lim_{M \rightarrow \infty} \text{Corr}_M(b_{i,t}, b_{j,t}) = 1.$$

The rank-one term $\Lambda_t \mathbf{1}\mathbf{1}'$ loads equally on all currencies via the common factor S_t . As systemic stress grows, this common component dwarfs all idiosyncratic and bilateral contributions, forcing all basis spreads to move together. Hence, extreme funding stress generates near-perfect cross-currency co-movement. $\square$

A.3. Proof of Proposition 2

Proof. Fix a currency k and consider a unit shock to its local demand $d_{k,t}$ with all other demands held fixed. Let e_k be the k th canonical basis vector and consider the perturbation

$$d_t \mapsto d_t + \varepsilon e_k, \quad \implies \quad \Delta d_t = \varepsilon e_k, \quad \varepsilon > 0$$

From the pricing relation $b_t = K_t d_t$, holding K_t constant, the resulting change in basis spreads

$$\Delta b_t = K_t \Delta d_t = \varepsilon K_t e_k, \quad (\Delta b_t)_j = \varepsilon K_{j,k}$$

Decomposing $K_t = T + \Phi + \Lambda_t 11'$, for $j \neq k$ we have

$$\Delta b_{j,t} = \varepsilon(\phi_{kj} + \Lambda_t)$$

since T is diagonal and $(\Lambda_t 11')_{j,k} = \Lambda_t$. Summing over $j \neq k$ gives aggregate outward spillover:

$$\sum_{j \neq k} \Delta b_{j,t} = \varepsilon C_k + \varepsilon(n-1)\Lambda_t, \quad C_k := \sum_{j \neq k} \phi_{kj}$$

Similarly, for a shock to another currency $i \neq k$:

$$\sum_{j \neq i} \Delta b_{j,t} = \varepsilon C_i + \varepsilon(n-1)\Lambda_t$$

Currency k is the central transmitter of shocks if the total impact on other currencies is larger than for any other $i \neq k$, i.e.

$$\sum_{j \neq k} \Delta b_{j,t} > \sum_{j \neq i} \Delta b_{j,t} \iff C_k > C_i$$

Hence, k is a central transmitter if and only if its centrality C_k is maximal among all currencies. □

A.4. Proof of Proposition 3

Assumption 4 (Structural Features of Dollar-Pegged Regimes). *The following stylized assumptions characterize tightly dollar-pegged currencies. They are not arbitrary constraints but institutional regularities observed in practice and documented by the BIS and prior literature (see Avdjiev et al., 2019a; Du et al., 2018a).*

- (i) **Zero net USD demand.** *Under a hard or tightly managed dollar peg, the domestic central bank effectively intermediates all foreign-exchange flows to maintain the peg. Consequently, private agents do not generate persistent net USD demand, implying*

$$d_{k,t} \approx 0.$$

In equilibrium, any USD liquidity imbalance is absorbed by the central bank's reserves rather than by the private dealer network.

- (ii) **Weak balance-sheet linkages.** *Pegged currencies typically exhibit limited bilateral funding and risk-transfer activity vis-à-vis non-dollar currencies due to capital-account restrictions, regulatory insulation, and policy alignment with the anchor currency (USD). Dealers and issuers in these markets therefore do not warehouse independent cross-currency exposures, yielding*

$$\sum_{j \neq k} \phi_{kj} \approx 0.$$

This captures the empirical observation that dollar-pegged currencies such as the HKD, SAR, and AED display near-zero cross-currency basis volatility and negligible network centrality.

The following proposition formally establishes that, under these institutional features, a tightly dollar-pegged currency exhibits lower basis volatility than any non-pegged currency.

Proof. Suppose currency k is tightly pegged to the USD so that (i) its own net demand is negligible, $d_{k,t} \approx 0$, and (ii) its balance-sheet linkages are small, $\sum_{j \neq k} \phi_{kj} \approx 0$ (isolation on the row of Φ) and (iii) let $d_t = \bar{d}_t \mathbf{1}$ (i.e., $d_{j,t} = \bar{d}_t$ for all j) or let $d_{j,t} = s D_t$ with the same share s for all j (equivalently $d_t = s D_t \mathbf{1}$), i.e. equal share of total USD demand from each currency $j = 1 \dots n$. From (A.1), the basis for currency k is

$$b_{k,t} = \tau_k d_{k,t} + \sum_{j \neq k} \phi_{kj} d_{j,t} + \Lambda_t S_t, \quad S_t = \mathbf{1}' d_t.$$

Under (i)–(ii), the first two terms are negligible, so

$$\text{Var}(b_{k,t}) = \text{Var}\left(\underbrace{\tau_k d_{k,t}}_{\approx 0} + \underbrace{\sum_{j \neq k} \phi_{kj} d_{j,t}}_{\approx 0} + \Lambda_t \sum_{j=1}^n d_{j,t}\right) \approx \text{Var}\left(\Lambda_t \sum_{j=1}^n d_{j,t}\right)$$

For any non-pegged currency i , define $X_{i,t} := \tau_i d_{i,t} + \sum_{j \neq i} \phi_{ij} d_{j,t}$ and $Y_t := \Lambda_t S_t$. Then

$$\text{Var}(b_{i,t}) = \text{Var}(X_{i,t} + Y_t) = \text{Var}(Y_t) + \text{Var}(X_{i,t}) + 2\text{Cov}(X_{i,t}, Y_t) \geq \text{Var}(Y_t)$$

with equality only if $X_{i,t} \equiv 0$ almost surely. Hence,

$$\text{Var}(b_{k,t}) \leq \text{Var}(b_{i,t})$$

demonstrating that the pegged currency's basis is less volatile than any non-pegged currency. This result follows from the idea that under the mild assumption that USD funding demands co-move non-negatively, $\text{Cov}(X_{i,t}, S_t) \geq 0$, and using that $\Lambda_t \geq 0$ is independent of d_t (hence independent of $X_{i,t}$ and S_t with $\mathbb{E}[S_t] = 0$), we have

$$\text{Cov}(X_{i,t}, Y_t) = \text{Cov}(X_{i,t}, \Lambda_t S_t) = \mathbb{E}[\Lambda_t] \text{Cov}(X_{i,t}, S_t) \geq 0.$$

Therefore,

$$\text{Var}(b_{i,t}) = \text{Var}(X_{i,t}) + \text{Var}(Y_t) + 2 \text{Cov}(X_{i,t}, Y_t) \geq \text{Var}(Y_t).$$

For a tightly pegged currency k , $b_{k,t} \approx Y_t$ (since $d_{k,t} \approx 0$ and $\sum_{j \neq k} \phi_{kj} \approx 0$), so $\text{Var}(b_{k,t}) \approx \text{Var}(Y_t)$. Hence,

$$\text{Var}(b_{i,t}) \geq \text{Var}(b_{k,t}),$$

which confirms that the pegged currency's basis is less volatile than that of any non-pegged currency. □

A.5. Proof of Proposition 4

Index tenors by $m \in \mathcal{M}$. For tenor m , the pricing equation is

$$b_t^{(m)} = K_t^{(m)} d_t^{(m)}, \quad K_t^{(m)} = T^{(m)} + \Phi^{(m)} + \Lambda_t \mathbf{1}\mathbf{1}',$$

with $T^{(m)} = \text{diag}(\tau_1^{(m)}, \dots, \tau_n^{(m)})$ and $\Phi^{(m)} = \Phi^{(m)'}.$ Assume:

(A1) $\tau_k^{(m_2)} \geq \tau_k^{(m_1)}$ for all k whenever $m_2 > m_1$.

(A2) $d_t^{(m)} = H_t(\tau^{(m)}) \varepsilon_t$, with $\mathbb{E}[\varepsilon_t] = 0$, $\Sigma_\varepsilon := \text{Var}(\varepsilon_t) \succeq 0$, and $H_t(\cdot)$ Loewner-decreasing componentwise in τ , i.e. if $\tilde{\tau} \geq \tau$ elementwise then $H_t(\tilde{\tau})H_t(\tilde{\tau})' \preceq H_t(\tau)H_t(\tau)'$.

(A3) ε_t is orthogonal to Λ_t .

(A4) For $m_2 > m_1$, there exists a contraction C_t with $\|C_t\|_2 \leq 1$ such that

$$K_t^{(m_2)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})' K_t^{(m_2)'} \preceq K_t^{(m_1)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})' K_t^{(m_1)'}$$

Proof. Conditional on Λ_t ,

$$\text{Var}(b_t^{(m)} \mid \Lambda_t) = K_t^{(m)} H_t(\tau^{(m)}) \Sigma_\varepsilon H_t(\tau^{(m)})' K_t^{(m)'} =: G_t^{(m)} \Sigma_\varepsilon G_t^{(m)'}$$

By (A1)–(A2) and the Loewner-decreasing property,

$$H_t(\tau^{(m_2)}) \Sigma_\varepsilon H_t(\tau^{(m_2)})' \preceq H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})'$$

Premultiplying and postmultiplying by $K_t^{(m_2)}$ preserves the order:

$$K_t^{(m_2)} H_t(\tau^{(m_2)}) \Sigma_\varepsilon H_t(\tau^{(m_2)})' K_t^{(m_2)'} \preceq K_t^{(m_2)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})' K_t^{(m_2)'}$$

By (A4), there exists a contraction C_t with $\|C_t\|_2 \leq 1$ such that

$$K_t^{(m_2)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon^{1/2} = C_t K_t^{(m_1)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon^{1/2}.$$

Hence,

$$\begin{aligned} K_t^{(m_2)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})' K_t^{(m_2)'} &= (K_t^{(m_2)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon^{1/2}) (K_t^{(m_2)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon^{1/2})' \\ &= (C_t K_t^{(m_1)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon^{1/2}) (C_t K_t^{(m_1)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon^{1/2})' \\ &= C_t (K_t^{(m_1)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})' K_t^{(m_1)'}) C_t' \\ &\preceq K_t^{(m_1)} H_t(\tau^{(m_1)}) \Sigma_\varepsilon H_t(\tau^{(m_1)})' K_t^{(m_1)'}, \end{aligned}$$

Thus,

$$\text{Var}(b_t^{(m_2)} \mid \Lambda_t) \preceq \text{Var}(b_t^{(m_1)} \mid \Lambda_t)$$

For spillovers, let $G_t^{(m)} = K_t^{(m)} H_t(\tau^{(m)})$:

$$\|(G_t^{(m_2)})_{k,-k}\|_2 \leq \|(G_t^{(m_2)})_{k,\bullet}\|_2 \leq \|G_t^{(m_2)}\|_2 \leq \|G_t^{(m_1)}\|_2$$

where the last inequality follows from the Loewner ordering and contraction property. Thus, longer-tenor spreads exhibit weaker cross-currency transmission. $\square$